

SPECIAL SECTION

Abstracts from the 33rd Annual Meeting

International Society for Developmental Psychobiology

New Orleans, LA

November 1–4, 2000

Abstracts:

International Society for Developmental Psychobiology, 33rd Annual Meeting

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TIMING IS EVERYTHING: ASSOCIATING MEMORIES ACROSS DELAYS IN 6-MONTH-OLDS. A. Vieira, R. Barr, & C. Rovee-Collier, Department of Psychology, Rutgers University, Piscataway, NJ 08854-8020, USA.

A time window is a period within which new information can be integrated with an existing memory. In prior research, infants learned to move a mobile by kicking and later were passively exposed to a physically dissimilar windchime. The windchime was integrated with the mobile memory whether exposed immediately after training or 4 days later, at the end of the time window. Moreover, exposing the windchime after the longer delay protracted retention by a week. Recently, the memory of a puppet imitation task, which lasts only 1 day, was protracted to 2 weeks by associating it with an operant train task that was remembered for 2 weeks. Presently, we asked if the train and puppet memories could be further protracted by associating them after longer delays. In Experiment 1, 6-month-olds learned to press a lever to activate a train. Seven days later an adult modeled three actions on the puppet in the mere presence of the train. Infants tested 7 days after that (14 days after training) remembered both tasks. In Experiment 2, infants again learned the train task; 14 days later, they received a 2-minute retention test with the train and, immediately afterward, the three actions were modeled on the puppet. Infants tested 7 or 14 days after the modeling (21 and 28 days after training) again remembered both tasks. Apparently, merely retrieving the train memory at the end of its time window (14 days) doubled its persistence with no additional training. Also, the puppet memory was associated with the retrieved memory of the train. Thereafter, the train and puppet tasks were remembered together, even though they were originally acquired 2 weeks apart. These results reveal that retrieval per se protracts retention and that subsequently encountered information can be integrated with a memory representation that has been retrieved. Presumably, this integration occurs while both events are co-active in short-term memory. This may be an important means of expanding the mnemonic network of immature organisms.

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INHIBITORY EFFECTS OF UNPAIRED US/CS PREEXPOSURE ON EYEBLINK CONDITIONING IN DEVELOPING RATS: ROLE OF CONTEXTUAL PROCESSING. A. N. Rush,¹ B. L. Robinette,² & M. E. Stanton,^{1,2*} Department of Psychology, UNC-Chapel Hill; ²Neurotoxicology Division. (MD-74B), NHEERL, ORD, US EPA, RTP, NC 27711.

Learned irrelevance (LIr) is a Pavlovian conditioning phenomenon in which unpaired preexposure to a CS and US retards subsequent paired conditioning involving these stimuli. In developmental studies of eyeblink conditioning in rats, LIr emerges between postnatal days (PND) 20 and 25 and continues to strengthen until PND 30 (Rush, Robinette & Stanton, *Dev. Psychobiol.*, in press). In order to examine the role the ontogeny of contextual processing may play in this effect, the present experiments asked if LIr on PND 30 is context-specific. Experiment 1 was a 2×2 factorial involving two preexposure conditions, unpaired US/CS presentation and chamber-only preexposure, and two context conditions, context-same and context-shift. Rat pups were surgically prepared and tested for eyeblink conditioning as described in Rush et al. (*ibid*). Training consisted of three 100-trial sessions on PND30, with rats receiving preexposure to either 50 unpaired trials or the chamber only, during the first half of the first session, and paired training during the second half and all subsequent sessions. Preexposure and subsequent training occurred in either the same context (context-same) or in different contexts (context-shift). Unpaired US/CS preexposure impaired acquisition relative to the chamber-only preexposure, confirming the presence of LIr. The shift in context did not alter this outcome. Experiment 2 showed that contextual fear conditioning did not generalize across the same and different fear conditioning did not generalize across the same and different contexts. Thus, failure to discriminate the two contexts cannot explain the results of Experiment 1. These findings indicate that LIr is not context-specific under the training conditions we used. Implications for brain and memory development are discussed.

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REPEATED ISOLATION STRESS OF THE NEONATE: DOPAMINE LEVELS AND VOCALIZATIONS FOLLOWING COCAINE ON DAY 10. K. Mallinson, G. Suresh, H. Sanchez, & P. Kehoe, Neuroscience Program, Trinity College, Hartford, CT, 06105, USA.

Repeated isolation stress of the infant rat has been found to produce immediate and enduring enhanced responsiveness to the psychostimulant amphetamine. Specifically, infant, juvenile, and adult animals that had experienced short, but multiple periods of isolation during the first week of life, demonstrated an enhanced behavioral and nucleus accumbens (NAc) dopamine (DA) response to amphetamine compared to nonisolated controls. The present study assessed the effects of cocaine administration in 10-day-old rat pups that were separated from the nest, dam, and siblings for 1 hr daily from postnatal days (PND) 2–9. In the first experiment, we tested ultrasonic vocalizations during 6 min of isolation following no injection, saline or cocaine (doses from 1.25–10 mg/kg). We found that PN 10 pups administered saline or no injection, vocalized significantly more if they had been isolated from PND 2–9 than nonisolated controls. Cocaine, at all doses, significantly suppressed vocalizations in both groups. In the second experiment, using in vivo microdialysis, we measured NAc DA levels of isolated and nonisolated rat pups given either ip cocaine (2.5–10 mg/kg) or saline. Using ice anesthesia, a CMA11 probe was surgically implanted followed by a 90-minute recovery period and a 1 hr baseline, after which cocaine was administered. No baseline differences in DA were seen between isolates and controls. Cocaine increased DA in all groups compared to saline; however, previously isolated pups had a significantly greater increase in NAc DA than nonisolates at all doses of cocaine. These data suggest that neonatal isolation stress potentiates isolation-induced vocalizations and produces a sensitized DA response to cocaine.

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NEONATAL RAT INGESTION OF SUCROSE: LEVELS OF NE AND DA USING IN VIVO MICRODIALYSIS. P. Kehoe, B. McConarty, G. Suresh, & K. Mallinson, Neuroscience Program, Trinity College, Hartford, CT, 06105, USA.

The infant rat model has been used to explore the development of CNS mechanisms that mediate ingestion of sweet tastants. Among other neurotransmitters, norepinephrine (NE) and dopamine (DA) have been implicated in the control of sucrose taste and ingestion. For the rat pup, intraoral ingestion requires using a cannula inserted into the tongue allowing the pup to actively ingest. Independent ingestion (adult-like feeding), lapping up the solution from the bottom of a container, is found as early as postnatal day (PND) 1. The present study examined NE and DA levels in the nucleus accumbens (Nac) of PND 10 rats, using in vivo microdialysis before, during, and after ingestion of either water, 10, or 20% sucrose solutions. A CMA-11 probe, and for some, an intraoral tongue cannula were surgically implanted during ice anesthesia followed by a 90-minute recovery and a 1 hr baseline, after which sucrose was either infused or offered. For both modes of ingestion, NE levels were significantly higher with 20% sucrose but did not vary from baseline with water, while 10% was intermediate. Interestingly, DA levels were elevated by both sucrose solutions but *only* with intraoral ingestion. DA levels of the Nac were unchanged by any of the solutions with independent ingestion. The data suggest that the neonatal Nac NE system is responsive to both infant and adult modes of sucrose ingestion whereas DA responds to intraoral ingestion alone.

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HORMONAL AND SOCIAL INFLUENCES ON SEX DIFFERENCES IN PARENTAL RESPONSIVENESS IN VIRGIN PRAIRIE VOLES. J. S. Lonstein, & G. J. De Vries, Center for Neuroendocrine Studies, University of Massachusetts, Amherst, MA 01003, USA.

Mated prairie voles (*Microtus ochrogaster*) form monogamous pairbonds and show similar biparental care to their pups. However, there is a sex difference in the parental responsiveness of adult virgins. Males are parental whereas females are not. Unlike many behavioral sex differences in other rodents, we found that testicular hormones during neonatal life are necessary for later parental responding in males, but that females' behavior cannot be masculinized by perinatal injections of testosterone. The ability of neonatal castration of feminize the behavior of virgin males also extends to their brain. We have demonstrated via in situ hybridization that neonatal castration of males completely feminizes later expression of vasopressin mRNA, a neuropeptide that is sexually dimorphic in unmanipulated rodents. Although perinatal androgen does not masculinize the behavior of females, their responses to pups can be modified by their social environment. If virgin females are not removed from the natal nest at weaning, but instead remain there until adulthood, they are as highly parental as males. This is true even if their sire and dam *do not have subsequent litters*. This facilitation of parental responding may be important since most virgin females do not leave the natal nest but instead remain there and provide alloparental care to younger siblings. Prairie voles, therefore, are a unique model to examine the factors controlling behavioral and neural development.

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MATERNAL ENVIRONMENT INFLUENCES NEUROENDOCRINE ACTIVITY IN PERIADOLESCENT GUINEA PIG PUPS. M. B. Hennessy, D. S. Maken, & F. C. Graves, Department of Psychology Wright State University, Dayton OH 45435, USA.

Guinea pigs are precocial mammals. Active maternal care is minimal and rarely occurs after about the first week. Yet previous results indicate the presence of the mother and associated stimuli influence the physiology and behavior of the young well beyond this age. Here, we examined the effects of the maternal environment on periadolescent ($\approx$ Day 40) offspring. Subjects were either kept continuously in the maternal cage or rehoused with only littermates for either 2 or 10 days prior to testing. The principal new findings of this study were that rehousing with littermates but not the mother resulted in higher plasma levels of testosterone in males, progesterone in females, and cortisol and ACTH in males and females, than did continuous housing with the mother until the time of testing. Progesterone, cortisol, and ACTH levels increased monotonically with the length of rehousing, indicating that these elevations were not responses to exposure to a novel cage, or related to emotional reactions to removal of the attachment object, per se. The results suggest that even in the periadolescent period, maternal stimuli can exert potent, unexpected influences on the neuroendocrine activity of the precocial guinea pig, and may influence psychobiological development at this time. Supported by NSF, IBN-9729595.

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THE EFFECTS OF ADOLESCENT ETOH EXPOSURE ON THE REINFORCING PROPERTIES OF ETOH DURING A SUBSEQUENT ADULT EXPOSURE. D. L. Graham, & J. L. Diaz-Granados,* Department of Psychology and Neuroscience, Baylor University; Waco, TX, 76706, USA.

Previous findings from our laboratory indicate that adolescent EtOH exposure alters the associative properties of EtOH. Specifically, our findings demonstrate that adolescent EtOH exposure attenuates the aversive properties of EtOH during a subsequent adult EtOH exposure. The primary purpose of the present experiment was to extend the above findings by examining the role of adolescent EtOH exposure on the *reinforcing* properties of EtOH during a subsequent adult exposure. Adolescent C3H mice (and appropriate controls) were exposed to 64 hr of continuous EtOH vapor on post-natal day (PND) 28. Approximately 6 weeks following adolescent exposure the rewarding properties of EtOH were examined using the conditioned place preference paradigm. During conditioning animals received four pairings of EtOH (2.5 g/kg) with a distinctive stimulus for a 5-minute session. On intervening days animals received a saline injection that was paired with a different stimulus. Place preference was tested by allowing animals access to both sides of the box for a 15-minute period. A place preference ratio was computed by dividing the total time spent in the EtOH-paired environment over the total time spent. Results indicate that C3H mice who were pre-exposed to EtOH during adolescence developed an EtOH place preference as compared to controls ($p < .01$). The findings from this study demonstrate that adolescent EtOH pre-exposure potentiates the reinforcing properties of EtOH during a subsequent adult exposure. The findings of the present study may have important implications with regard to the effect of adolescent alcohol abuse on alcohol-related behaviors in adulthood.

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DEVELOPMENT OF ANXIETY-LIKE RESPONDING IN THE ELEVATED PLUS MAZE. S. L. Andersen, & A. Gee, McLean Hospital and Harvard Medical School 115 Mill Street, Belmont, MA 02148, USA.

Collectively, anxiety disorders are the most common psychiatric condition in children. Despite its prevalence, anxiety during development has received very little attention at the level of basic research. To assess how anxiety changes across age, rats the ages of 15, 21, 25, 30, 40, 60 and adult (>150) days were tested in an elevated plus maze. Subjects ($n = 6 = 8/\text{age and sex}$) were placed in the maze facing the closed, and the time spent in the open versus closed arms was recorded for a 5-minute period. The maze was then washed out with a mild disinfectant between subjects to minimize odor cues. The percentage of time spent in the closed arm (indicative of increased anxiety) increased significantly with age ($F_{6, 65} = 7.14, p < .0001$). The sex of the subjects did not influence maze performance ($p > .1$). To determine if age-related differences in activity influenced anxiety-like responding, the time spent in the closed arm was corrected for the total number of crossovers between the arms. Age had a significant influence on this parameter as well ($F_{6, 53} = 16.82, p < .0001$). Post hoc tests with Bonferroni correction (at the $p < .05$) revealed significant differences between 15- and 21-day-olds when compared with the other ages. However, even though these ages were the most active, their level of anxiety still remained lower than the other ages. Future studies from our laboratory will investigate the developing neurochemistry that underlies these effects. We have recently shown that anxiety-like behavior is highly correlated ($r = .85$) with the lateralization of serotonin in the amygdala and dopamine in the prefrontal cortex in adults. Future studies will determine how the laterality of these key neurotransmitters change with development.

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DOPAMINE SYNTHESIS, TURNOVER, AND METABOLISM IN YOUNG AND ADULT RATS: EFFECTS OF CCK_A RECEPTOR ANTAGONISM. D. L. Abbott, A. R. Zavala, A. Nazarian, H. De La Rosa, W. A. Brinkley, S. A. McDougall, & C. A. Crawford, Department of Psychology, California State University, San Bernardino, CA 92407, USA.

Previous research has suggested that the cholecystokinin (CCK) system has behavioral impact by modulating the functioning of the dopamine (DA) system. The purpose of this study was to determine whether a CCK_A receptor antagonist (lorglumide) would alter striatal DA synthesis, turnover, and metabolism. To this end, 14- and 90-day-old rats were injected with saline or lorglumide (1.0 mg/kg, ip) followed, 20 min later, by an injection of saline or amphetamine (1.0 mg/kg, ip). After 20 min, rats were given either two injections of saline or an injection of the DOPA decarboxylase inhibitor NSD-1015 (100 mg/kg, ip) and the MAO inhibitor pargyline (75 mg/kg, ip). Brains were removed 30 min later and stored at -80°C . L-DOPA, DOPAC, and DA were measured using HPLC-EC. DA synthesis was estimated by measuring DOPA levels in NSD-treated rats. DA turnover was estimated by measuring DOPAC efflux and by comparing the ratio of DOPAC to DA. DOPAC levels were used as a measure of DA metabolism. Results showed that amphetamine increased DA synthesis in both 14- and 90-day-old rats. Amphetamine also decreased DA turnover and metabolism in both age groups. Contrary to expectations, a lorglumide did not affect baseline measures of DA synthesis, turnover, or metabolism, nor did lorglumide modulate amphetamine's actions. Therefore, these results show that amphetamine affects striatal DA synthesis, turnover, and metabolism similarly in both young and adult rats: actions that were unaffected by CCK_A receptor antagonism.

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EFFECTS OF WARMTH ON FEEDING IN HUMAN INFANTS. L. Bacher, L. Tellefsen, R. Romm, N. Korba, S. Robertson, & W. Smotherman, Department of Psychology, Binghamton University, Binghamton, NY, USA.

Warmth is a powerful regulator of behavior in young animals. Previous research indicated that warming newborn rats resulted in reduced frequency of oral grasping of an artificial nipple, yet the grasps were significantly longer over time. Our objective was to examine the effects of warmth on feeding in human infants. Twelve, 10–12-week-old, bottle-fed babies were observed before, during, and after a feeding. Half of the infants were warmed throughout the observation by a small heating pad in the seat in which the infants sat. Videotapes of the infants were used to determine sucking rates and the amount of time infants sucked. At each break in the feeding, the bottle was weighed to estimate the amount of milk consumed. Warmed infants spent less time on the bottle ($p = .03$), drank less milk ($p = .05$), and sucked fewer times ($p = .02$) than non-warmed babies. However, warmed babies may have consumed more milk per suck and therefore, may have sucked more efficiently.

Being able to alter aspects of feeding through a simple manipulation of temperature may suggest ways to promote feeding in infants who are not feeding well.

Research supported by MERIT Award HD 16102 to WPS.

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MATERNAL CARE AND OFFSPRING BEHAVIOR IN 5-HT_{2A} RECEPTOR MUTANT MICE. M. Bagnall, E. Marcianno, D. Leven, & J. Gingrich, Department of Developmental Psychobiology, NYS Psychiatric Inst. New York, NY, 10032, USA.

High levels of maternal care in rats have been shown to correlate with reduced anxiety levels in offspring, in a range of behavioral and physiological tests (Liu, D. et al., 1997). In this study it is shown that 5-HT_{2A} receptor knockout (KO) mice display lower levels of maternal care than wild-type (WT) mice, as characterized by their levels of arched-back nursing. KO offspring in turn explored less in the open field than wild-type offspring, though their performance in the elevated-plus maze was identical to the wild-type behavior. In contrast, when dams were bred to opposite-genotype males (KO dams to WT males, WT dams to KO males), KO dams displayed the same levels of care to their heterozygous offspring as WT dams did, suggesting that the genotype of the offspring has a significant effect in inducing maternal behavior. The heterozygous offspring of WT dams display slightly elevated ultrasonic vocalization at 8 days of age, which may either affect or be affected by levels of maternal care.

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SENSORY PRECONDITIONING AND DEFERRED IMITATION IN HUMAN INFANTS. R. Barr, H. Marrott, & C. Rovee-Collier, Rutgers University, Department of Psychology, Piscataway, NJ 08854-8020, USA.

Six-month-olds will not imitate modeled actions on any test object except the original one. We asked if infants who associated the two objects in a sensory preconditioning procedure would do so. To answer this, *group paired/B* was exposed to two hand-puppets (A, B), side-by-side, 1 h/day for 7 days. On Day 8, the target actions were modeled on puppet A, and infants were tested 24 h later with puppet B. *Group paired/C* was treated identically but was tested for generalization with a completely novel puppet. There were three control groups—*group unpaired/B* was preexposed equally long to A and B at different times of day; *group no preexposure* was simply trained with A and tested with B; and a *baseline control group* was only tested with B. Only *group paired/B* imitated the target actions on B, suggesting that an association had formed between puppets A and B during their paired preexposure. The failure of *group paired/C* to imitate confirmed that the imitation by *group paired/B* was not due to stimulus generalization. In Experiment 2, *group paired/B-train* learned an operant train task on days 8 and 9 after their paired exposure to puppets A and B. On Day 9 demonstration of the target actions on puppet A occurred in the context of the train set—a procedure known to associate puppet A with the train. After both the train and puppet tasks were forgotten, infants were reminded for 30 sec with puppet A. They were tested for retention 24 hr later first with the train and then with puppet B. *Group paired/C-train* was treated identically and was tested with the train and puppet C. The *puppet-only group* did not learn the train task but was preexposed, reactivated, and tested with puppet B. Puppet A reactivated the memory of the train task for *groups paired/B-train* and *paired/C-train*, but only *group paired/B-train* imitated the target actions. Failure of the *puppet-only group* to imitate the target actions revealed that imitation by *group paired/B-train* depended on associating puppet A with the train. We conclude that the reminder (puppet A) reactivated two associations—the association between puppet A and the train, and the association initially formed between puppet A and puppet B. These data show that sensory preconditioning effects are robust and can form the basis for later complex behavior.

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INTERSENSORY REDUNDANCY: INFANT DETECTION OF UNIMODAL PROPERTIES OF EVENTS. L. E. Bahrick, R. Lickliter, & R. Flom, Department of Psychology, Florida International University, Miami, FL 33199, USA.

Bahrick and Lickliter (2000) proposed an intersensory redundancy hypothesis which predicts that in early development information presented redundantly across two sensory modalities recruits attention to the redundantly specified property at the expense of other properties. Second, when the same event is presented to one sense modality alone attention is recruited to the unimodal aspects of the event and facilitates perceptual learning of unimodal properties at the expense of other properties. Prior research has supported the first prediction of this hypothesis, but to date no test of the second prediction has been conducted. This experiment investigated the second prediction by examining infants' detection of orientation, a unimodal visual property under both unimodal and bimodal presentation conditions. Five-month-old infants ($N = 32$) were habituated to films of a hammer tapping out a rhythm in one of two orientations (up vs. down) under either a unimodal (visual) or a bimodal (auditory and visual) condition. Results confirmed our hypothesis and indicated that following unimodal habituation infants showed significant visual recovery to the change in orientation ($p < .02$), but following bimodal habituation infants' visual recovery did not reach significance. These findings, with prior results, provide support for both predictions of the intersensory redundancy hypothesis and have important implications for the development of perception.

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EFFECTS OF OLFACTORY DEPRIVATION ON LAYER III INTERNEURONS IN RAT ANTERIOR PIRIFORM CORTEX. A. R. Best, & D. A. Wilson, Department of Zoology, University of Oklahoma, Norman, OK, 73019, USA.

Experience early in life can result in changes in adult neural structure. There is increasing evidence that interneurons play a critical role in neocortical function, controlling excitability, stimulus-induced electrical oscillations and serving as targets for modulatory inputs. Interneurons serve a similar role in the piriform cortex; thus we were interested in how early experience affects growth and connectivity of this heterogeneous class of cortical neurons. Layer III interneurons were reconstructed from anterior piriform cortex of sham-operated and unilaterally nares-occluded rats. The 29–31 days of nares occlusion began either on PND1–2 or PND30. On PND29–31 or PND59–61, biotinylated dextran was ionophoresed into piriform Layer II/III. Animals were killed after 4–5 days survival post-injection and tissue was processed by the ABC method. Reconstructions were performed on a camera lucida at 400 $\times$ and 1000 $\times$ magnification. Cells were then reconstructed in three dimensions using metamorph software. Measures of dendritic complexity and extent will be used to compare between groups. The relatively small number of interneuron types and simple laminar organization of piriform cortex may allow new insight into the development of interneuronal microcircuits in other cortical areas. Supported by NSF grant IBN-9808149.

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BEHAVIORAL THERMOREGULATION IN INFANT RATS AND HAMSTERS. M. S. Blumberg, G. Sokoloff, E. A. Boline, & N. M. Streeper, Department of Psychology, University of Iowa, Iowa City, IA, 52242, USA.

The ability to orient and locomote toward a warm stimulus is of paramount importance to altricial infant mammals. Whereas previous investigators have used a variety of techniques to assess behavioral thermoregulation, we have developed a single methodology for studying infant rodents of different ages and species. The apparatus consists of a 6×10 array of peltier diode "tiles" (4×4 cm) in which each tile's surface temperature can be independently controlled. The temperature of each tile and subject is monitored using infrared thermography. Subjects are placed on four central tiles initially maintained at a warm temperature (39°C) and are confined within a lightweight ring. Tiles outside of the ring are also maintained at 39°C . When the temperature of the four central tiles is decreased, we measure whether the subject escapes from the cool central region. For this study, infant rats and hamsters at 2 and 8 days of age were used; these two species differ in that infant rats can produce heat endogenously whereas infant hamsters cannot. In three experiments, we assessed infants' sensitivities to varying levels of cooling and whether they use thermal information outside of the ring to guide their behavior. We also asked whether pharmacological blockade of endogenous heat production in rats improves their performance. Our results provide evidence of thermotaxis and thermokinesis at different ages in these species and lay a foundation for future studies of the sensory and neural mechanisms of behavioral thermoregulation in infants.

This research was supported by NIMH grant MH50701.

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THERMOGRAPHIC STUDY OF BROWN ADIPOSE TISSUE ACTIVATION IN INFANT ICR MICE. T. E. Brine, & J. R. Alberts, Department of Psychology, Indiana University, Bloomington, IN 47405, USA.

Most studies of thermogenesis by brown adipose tissue (BAT) imply, by virtue of the experimental design and parameters, that BAT thermogenesis is a mechanism used in thermal "emergencies" as a final effort to protect body temperature. We exposed mouse pups to roughly sinusoidal fluctuations in ambient temperature (max range: $37\text{--}26^\circ\text{C}$). Pups repeatedly activated and de-activated interscapular BAT thermogenesis as ambient temperature decreased and increased. Infrared thermography enabled us to visualize when and for how long pups of different ages are thermogenic. We suggest that BAT thermogenesis is not confined to "emergency" situations, but may be predominantly used when pups face more modest, normally occurring variations in ambient temperature. Indeed, pups may use BAT to regulate body temperatures while in the nest. We compared the BAT activity of single pups to that of a group of age mates. BAT thermogenesis is displayed by pups in huddles.

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BEHAVIORAL RESPONSES TO THERMAL STIMULATION IN INFANT RATS AND HAMSTERS. E. A. Boline, G. Sokoloff, & M. S. Blumberg, Department of Psychology, University of Iowa, Iowa City, IA, 52242, USA.

Successful behavioral thermoregulation in infants exposed to cold requires both orientation and locomotor responses. In this study, we examine the locomotor responses of 2- and 8-day-old Norway rats and Syrian golden hamsters during thermal stimulation. Because these two species differ in that infant rats can produce heat endogenously whereas infant hamsters cannot, they can be used to assess the contributions of endogenous heat production to behavioral responding in the cold. For this experiment, we used a novel apparatus consisting of a 4×4 array of peltier diode "tiles" (4×4 cm). Tile and pup skin temperatures are monitored using infrared thermography. Initially, a pup is acclimated on warm tiles maintained at a temperature of 39°C , whereupon the pup falls asleep. During the 120-minute test, the temperature of all 16 tiles is decreased 4°C at 30-minute intervals, 8°C at 60-minute intervals, or left unchanged for 120 min. Because all the tiles are maintained at the same temperature, cues for thermal orientation are not available to the pups. Infrared images are acquired every 10 s throughout the test and the pup's behavior is recorded to videotape. Locomotor responses (e.g., square crosses) and dorsal skin temperature are analyzed. The design employed here allows us to determine the relative contributions of absolute body temperature and change in body temperature to behavioral activation, as well as examine developmental and species differences in behavioral thermoregulation.

This research was supported by NIMH grant MH50701.

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EFFECTS OF UNILATERAL FORELIMB WEIGHTING ON SPONTANEOUS MOTOR ACTIVITY IN NEONATAL RATS. M. R. Brumley, & S. R. Robinson, Department of Psychology, University of Iowa, Iowa City, IA, 52242, USA.

Experiments in which an external weight was applied unilaterally to the leg of a human infant have suggested that proprioception plays an important role in the early coordination of spontaneous limb movement. Some studies have found that applying a weight to one leg affects the behavior of both the weighted and unweighted legs, implying the existence of intrinsic within-girdle constraints on limb activity. Perhaps because humans employ a bipedal gait as their primary locomotor pattern, previous studies of human infants have focused exclusively on leg activity. However, quadrupedal locomotion is typical of most other mammals, and this study therefore sought to examine whether a weight applied to one limb would affect the amount or patterning of activity expressed by all four limbs. We hypothesized that neonatal rats would alter the organization of spontaneous limb movements as they adjusted to changes in proprioception induced by application or removal of a limb weight. Each weight was calibrated to approximate 0, 25, 50, or 100% of the average mass of one forelimb, and was attached around the circumference of the limb proximal to the forepaw. Changes in the frequency of movement of both forelimbs and hindlimbs were evident during a 30-minute period when the weight was attached. In subjects exposed to 50 or 100% limb weights, the unweighted forelimb showed a more pronounced change in activity than the weighted limb, and changes in activity persisted during the 30-minute period after the weight was removed. These findings indicate that changes in proprioceptive feedback may play an important role in the organization of spontaneous limb activity in the neonatal rat, and by extension, in other developing mammals.

This research was supported by NIH grant HD 33862 to SRR.

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DEVELOPMENTAL EFFECTS OF SELECTIVE BREEDING FOR AN INFANTILE TRAIT: RAT PUP ULTRASONIC VOCALIZATION (USV). S. A. Brunelli, M. A. Hofer, J. R. Masmela, & H. N. Shair, Department of Developmental Psychobiology, NYS Psychiatric Inst. New York, NY, 10032, USA.

This study asks two questions: (1) How 14 generations of selective breeding or rats for high or low levels of USV response at 10 days of age may affect the time course of USV response development from 3 to 21 days of age; and (2) how the development of other behavioral and physiological traits may have been altered over the same period in association with changes in the selected trait. We conducted a combined longitudinal and cross-sectional study of rat pups from a total of 80 litters of high, low, and random lines. Two cohorts of litters were tested: one at PND 3, 10, and 18 age and the second at PND 7, 14, and 21, to minimize possible repeated testing effects. In addition to the USV response to 2-minute isolation tests, other isolation-induced behaviors, physical markers of maturation, and measures of thermoregulation, were studied.

The results support the idea that the developmental trajectories of high and low lines have been altered differently, that is, that developmental paths are independent of each other. These results will be interpreted based on hypotheses about effects of continued selection on developmental processes.

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INTRANASAL ZINC SULFATE SPRAY INDUCES ANOSMIA BUT DOES NOT FACILITATE MATERNAL BEHAVIOR (MB) IN ESTROUS RABBITS. ¹R. Chirino, ¹C. Beyer, ²J. S. Rosenblatt, & ¹G. González-Mariscal, ¹CIRA, CINVESTAV-UAT, Tlaxcala 90000, Mexico; ²Department Psychology, Rutgers University, Newark, NJ 07102, USA.

Bilateral lesions of the accessory olfactory bulbs facilitate MB in virgin rabbits daily exposed to pups, a finding suggesting that the accessory olfactory system tonically inhibits such behavior in non-lactating females. To investigate if the main olfactory system exerts a similar inhibitory influence we lesioned the olfactory epithelium by spraying ZnSO₄ (5% solution) into the nasal cavity of virgin adult rabbits. A test was performed to determine a normal olfactory perception before infusion and to verify ZnSO₄-induced anosmia. Rabbits were deprived of food for 24 hr and then given dog chow as the single food source: only females that refused to eat it before the olfactory lesion but began eating it from then onward were included in the experimental group. Control females were those sprayed with saline that consistently refused to eat dog chow. Olfactory perception was determined in all animals every 4th day throughout the experiment. Animal well-being was verified on alternate days by quantifying: rabbit chow ingestion, scent-marking, and ambulation in an open field. MB was determined in all females by exposing them to four pups (1–7 days old) for 1 hr across 14 consecutive days. All females ($n = 6$ experimental; $n = 3$ control) refused to enter the nest box, placed inside their home cage and containing the pups. Three anosmic females showed aggression toward the litter and cannibalism. Results suggest that, in rabbits, the main and the accessory olfactory systems play different roles in modulating the expression of MB in virgin rabbits.

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PERCEPTUAL CONFIGURATION IN NEWBORN RATS: SENSORY PRECONDITIONING AND SECOND ORDER CONDITIONING. S. J. Cheslock, J. M. High, E. I. Varlinskaya, & N. E. Spear, Department of Psychology, Binghamton University, NY, 13902, USA.

As a general rule, adults outperform infants on learning and memory tasks; however, in some instances infants display superior performance. A trend has emerged suggesting that infants gain superiority when the task demands stimulus configuration (e.g., Kucharski & Spear, 1985; Chen et al., 1991), suggesting an infant predisposition to perceptually configure stimuli. The present study shows strikingly robust instances of “higher order” conditioning—sensory preconditioning (SP) and second-order conditioning (SC)—in naive, newborn rats. Both SP and SC involve two phases of conditioning, one in which two stimuli (e.g., odors) are paired, forming a compound (AB), and another in which the value of only one member of the compound (A) is altered in some way—by pairing with footshock, for example. Conditioning is evident when the other stimulus (B) acquires the new value of the altered stimulus (A). Infant rats in their second and third weeks of life are remarkably skilled at these tasks and outperform adults, but only when the stimuli are presented simultaneously; sequential presentation obliterates the infant's advantage (Chen et al., 1991). The present study employed a classical conditioning paradigm, in which newborn rats have shown extremely robust first-order conditioning, even with as much as a 60 second trace interval imposed between the conditioned and unconditioned stimuli (Cheslock et al., 2000). In the present experiment newborns showed very robust SP and SC, but only if the stimuli to-be-associated occurred simultaneously; no association was acquired with sequential presentation, even with a 0-second interstimulus interval. These data lend support to perceptual configuration as the encoding strategy in infancy, from the beginning of postnatal life.

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USE OF THE CPRS-48 WITH CHILDREN UNDER 3 YEARS OF AGE. A. D. Clements, Human Development and Learning Department, East Tennessee State University, Johnson City, TN 37614, USA.

The Connors Parent Rating Scale-48 (CPRS-48) has consistently been shown to be a valid indicator of hyperactivity, impulsivity, conduct disorder, learning problems, anxiety, and psychosomatic tendencies in children aged 3 years or over. In my current line of research, the investigation of prenatal events' contributions to hyperactivity and impulsivity in children, I am interested in having a measure that will discriminate hyperactivity and impulsivity at a younger age. This study compared the CPRS-48 ratings of a group of children over 3 years of age (mean age = 6.63, range 36 months–13 years; and a group of children younger than 3 years (mean age = 1.61, range 5–35 months). Using MANOVA, it was found that there were no significant differences between children younger than 3 years and those older than 3 years on any of the subscales. A χ^2 Test of Independence was also performed for the number of children with T-scores above 50 on each subscale for children aged 3 years or older and for children under 3 years ($\chi^2 = 0.043$, n.s.). The CPRS-48 may have promise as a behavior screen at younger children that was previously used in validation studies.

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BEHAVIORAL RESPONSES TO ACUTE AND CHRONIC NICOTINE EXPOSURE IN INFANT RATS. I. Cohen, & C. P. Cramer, Department of Psychological and Brain Sciences, Dartmouth College, Hanover, NH 03755.

The effects of prenatal exposure to nicotine, as well as tolerance to and withdrawal from nicotine in adults, have been studied extensively; however, the specific effects of nicotine during infancy have received considerably less attention. In the current study, a paw-lift latency test and an open field test were used to evaluate the effects of acute or chronic exposure to nicotine on rat pups at various ages during the first 3 postnatal weeks. An acute effect of nicotine on analgesic response first appeared at PND 7; however, there was no evidence for tolerance at this age following chronic exposure. Tolerance to the analgesic effect of nicotine was demonstrated at Day 17, as were both acute and chronic effects of nicotine on several open-field behaviors. There was, however, no evidence of antagonist- (mecamylamine) precipitated withdrawal in the open field. These results will be discussed in terms of both the ontogeny of nicotinic receptors and comparable patterns of emergence of tolerance to other psychoactive drugs.

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1.5-G HYPERGRAVITY EXPOSURE DURING DEVELOPMENT IMPAIRS RIGHTING RESPONSES OF PREWEANLING RATS. M. M. Conrad¹, E. A. Roldan,² & A. E. Ronca,³ ¹San Jose State, University, San Jose, CA 95192, USA, ²DeAnza College, Cupertino, CA 95014, USA, ³NASA Ames Research Center, Moffett Field, CA 94035, USA.

The righting response, movement of the body from supine to prone, is an indication of an animal's ability to orient itself with respect to the gravity vector. Righting is present at birth but continues to develop postnatally, with various forms of righting maturing at different rates (Pellis, Pellis, & Teitelbaum, 1992). Previous studies indicate that reducing gravitational input to the developing vestibular system impairs righting responses (Ronca & Alberts, 1997). Deficits in water-immersion righting responses were observed in neonatal rats that underwent a portion of their prenatal development in space. In the present study, we hypothesized that exposure to increased or hypergravity during development would augment postnatal righting responses. Rat offspring were exposed to continuous, 1.5g (16 rpm) centrifugation from Gestational day (G) 11 until one of three postnatal (P) ages: P6, P15, or P21. Righting responses were videotaped at high speed (1/1000 s) for subsequent analysis during slow-motion playback. Contact righting did not differ across the three age groups. Air-righting, tested on PND15 and PND21, was significantly poorer in 1.5 g-reared p15 rats relative to 1.0 g stationary controls. Righting ability was comparable for the PND21 rats reared at 1.5 and 1.0 g. The results suggest that gravitational input greater than that experienced on earth may be detrimental to the developing vestibular system. *Supported by NASA Grant 121-10-40.*

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STABILITY OF HEART RATE-BLOOD PRESSURE PRODUCT AN INDEX OF CARDIOVASCULAR RESPONSES TO FEEDING IN NEWBORN INFANTS. M. Cohen, M. M. Myers, D. R. Brown, & L. Solecki, Division of Neonatology, Children's Hospital of New Jersey at Newark, Beth Israel Hospital, Newark, NJ, ZIP, USA.

The young of many species exhibit increases in systolic blood pressure (SBP) during the first few minutes of feeding. In bottle feeding infants we found effects of volume of intake and prior pacifier experience on the magnitude of this response, but no effects of age, state, type of nipple, or taste. However, state had a great effect of heart rate (HR) changes to feeding, with larger HR increases in sleeping versus awake infants. Our most recent studies indicate that, independent of these systematic variations in feeding conditions, the magnitude of the SBP response was much smaller than in prior studies ($\approx +3$ mmHg vs. $\approx +8$ mmHg, $p < .05$). In contrast, HR responses to feeding have become larger across these studies ($\approx +7$ bpm vs. $\approx +13$ bpm, $p < .05$). These differences were primarily due to changes in baseline values prior to feeding and not to a change in values during feeding. This prompted an inquiry of the possible use of the product of SBP $\times$ HR (BPHR) as a more stable index of feeding responses. This rate-pressure product has been used as an indirect index of cardiac work and has been shown to correlate with coronary blood flow. Analyses of data collected from several studies show that baseline and feeding values of BPHR, as well as changes in BPHR in response to feeding, are quite invariant. While state influenced baseline levels of BPHR, changes in BPHR in response to feeding are relatively unaffected by differing sleep states. We thus suggest that a certain level of cardiac work is required to support the effort of feeding and the circulation adjustments occurring in preparation of nutrient processing. The BPHR product provides a better index of these physiologic responses than BP or HR alone.

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EFFECTS OF POSTURE AND TASK CONSTRAINTS IN HAND PREFERENCE ON HUMAN INFANTS. D. Corbetta, & E. Wilson, Departments of HKLS and Biological Sciences, Purdue University, West Lafayette, IN 47907, USA.

Human infants, like non-human primates, display weak biases in hand preference. Studies in non-human primates, however, have demonstrated that population-specific lateral biases can be revealed if the tasks are more demanding or if the subjects are tested in an upright, rather than quadruped posture. This study examined whether human infants' biases in hand preference can similarly be revealed by greater task constraints, and modulated by posture as infants learn to crawl on four limbs and later walk upright. Forty infants aged 7, 9, 11 and 13 months were asked to reach for objects at midline, in three different conditions: simple reaching of objects at midline, or reaching for the same objects in a wide or narrow container at midline. Most 7-month-olds were non-crawlers or had only a few weeks of crawling experience. The 9- and 11-month-olds were all confirmed crawlers, and the 13-month-olds were all walking independently for at least 2 months. Results showed that hand preference in simple reaching was the strongest in the 13-month-olds, and that hand preference tended to increase with age. Reaching in the containers tended to augment these lateral tendencies, but differences in hand preference between reaching conditions were significant only in the 7- and 13-month-old groups; that is when infants were not crawling on four limbs. These results indicate that task constraints help reveal lateral biases in human infants, just like in non-human primates. However, this effect is modulated by the different postures used by the infants at the different developmental ages. These results have potential implications for the developmental origins of hand preference and for human evolution.

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A FAR-FETCHED MATERNAL EFFECT. J. B. Craven, K. Linn, S. Snow, A. McKnight, C. Lucas, S. Benford, K. Grogan, J. Evans, H. L. Schreiber, C. Schreiber, & H. Schreiber, Department of Psychology, University of Texas, Tyler, TX 75799, USA.

This experiment showed that disturbing a mother rat's first litter influences the adult behavior of her second unmanipulated litter. The experiment had two phases. Dams and first litters were manipulated in the first phase. In the NESTCAGE group, nestcages with dams and litters inside were removed to areas remote from the colony room for 20 min daily for the first 8 days after birth (Day 0). CONTROL nestcages were left undisturbed in the colony room. In the second phase, dams and second litters, an age-matched PRIMIP control group, were not disturbed. Three males from each second phase litter were group-housed after weaning. The heaviest and lightest males were tested. Adult, male, second-phase offspring were confronted with a large, unfamiliar, caged, adult male in an open field, after they had been extensively habituated to the open field. There were no differences among the heavier animals, but, among the lighter siblings, CONTROL and PRIMIP animals spent twice as much time in contact with the unfamiliar adult male's cage as NESTCAGE animals, $t(20) = 2.915$ and $t(20) = 4.676$, respectively, $p < .01$. A similar "nestcage/parity" effect was found in a previous study (ISDP meeting, 1997), but these results should be regarded as tentative until replicated. [UT Tyler intramural grant; hschreib@mail.utyi.edu]

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NEURAL CORRELATES OF AN INHIBITORY CONTROL TASK. E. P. Davis, J. Bruce, K. Snyder, & C. A. Nelson, University of Minnesota, Minneapolis, MN, USA.

Inhibitory control is the ability to block a prepotent response. Studies of inhibitory control have found that areas of the frontal cortex play a role in response inhibition. This study was designed to examine the task-related processes of an inhibitory control task that had previously been found to involve activation of the dorsolateral and orbitofrontal areas in fMRI studies (Casey et al., 1997). Event-related potentials were recorded from 22 6-year-old children and 20 adults during a Go, No-Go task that contained a control and a response inhibition condition. We examined whether amplitude, latency, and topography of two components (the N2 which has been found to reflect attention and the P3 which has been found to be greater to task-relevant stimuli) were modulated by task condition. Both adults and children displayed longer reaction times and a greater number of errors in the response inhibition condition ($p < .01$). Greater negativity at the N2 was seen over frontal as compared to parietal leads for both adults ($p < .05$) and children ($p < .01$). However, for children, this effect is mediated by task condition with greater negativity over frontal leads seen for the response inhibition, but not control, condition ($p < .01$). Latency to the peak of the N2 was greater during the response inhibition condition for children ($p < .01$). Additionally, both adults ($p < .01$) and children ($p < .01$) showed greater amplitude of the P3 in the response inhibition condition, but not a longer latency to peak. ERP components were also associated with behavioral performance. In summary, the difference in pattern of activation between control and response inhibition conditions on this Go, No-Go task suggests that these conditions reflect different underlying processes. It seems that the response inhibition condition requires frontal attentional processes.

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A MOUSE MODEL OF GLUTARIC ACIDEMIA. L. S. Crnic, D. M. Koeller, M. Wootner, & S. I. Goodman, University of Colorado School of Medicine, Denver, CO 80262 USA.

Glutaric acidemia is caused by mutation in the gene for glutaryl CoA dehydrogenase (GCDH) and is an unusual inborn error of metabolism in that the majority of damage to the CNS occurs precipitously (over 48 hr) during a metabolic crisis in the early years of life, often apparently triggered by a virus infection. In addition, the damage is restricted to the basal ganglia. If such a metabolic crisis can be avoided or supportively treated, the outcome is good. The disorder is so rare that studies of the etiology and treatment of the neuron loss are difficult; thus, we constructed a mouse model for the disorder by inactivating GCDH by targeted mutagenesis. The resulting mice (whether on a hybrid or inbred background) displayed no GCDH enzyme activity in any tissue, and, similar to humans, showed elevated levels of glutaric acid in urine, serum, and CSF and have fatty livers. There was, however, no evidence of neuron loss in the basal ganglia and the motor symptoms of the disorder were very mild, with no significant differences in postnatal growth and motor development. The mice had mild, but significant reductions in performance on the accelerating rotarod, reaching significantly lower peak speeds in a 5-day training paradigm. They took significantly longer on their first attempts to traverse a narrow beam, but reached normal performance with training and showed no deficits on prepulse inhibition of the auditory startle response, a task known to be sensitive to basal ganglia damage. Current efforts to mimic the metabolic crisis that precipitates symptomatology in humans include exposure to inducers of cytokines (poly I:C and lipopolysaccharide), killed and live virus infection, food deprivation, and chilling. Supported by HD04024 and HD08315.

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DISSOCIATING THE IMPACT OF PRENATAL EXPOSURE TO THE PHONOLOGICAL AND PROSODIC FEATURES OF MATERNAL SPEECH. A. DeCasper, P. Prescott, F. Davis, & A. Elkins, Psychology Department, UNC-Greensboro, Greensboro, NC 27416, USA.

The phonological and prosodic properties of speech occur whenever a pregnant woman speaks. Probably, more of the prosodic features reach the fetus but the relative impact of each class of cues on perceptual development is unknown. In order to discern which properties of prenatal maternal speech might influence the newborn's operant learning, we experimentally manipulated the auditory reinforcer used. Whether it was input to the left or right ear, and the nature of the operant learning task. In simple tasks with English prose spoken by an unfamiliar female as the reinforcer, learning occurred *only* with input to the right ear (phonological features mediated by left side substrates). With intrauterine heartbeat sounds as the reinforcer, learning occurred *only* with input to the left ear (linguistically irrelevant prosodic cues mediated by right side substrates). However, in preference tasks pitting maternal voice reinforcers against non-maternal voice reinforcers (language held constant) and English prose against foreign language prose (voice held constant), the well-documented preferences of the newborn for the maternal voice and native language occurred *only* with input to the left ear (right hemisphere). As conjectured, these preferences seem to be based on prosodic cues. We conclude that both the phonological and prosodic properties of maternal speech impact developing fetal perception, they affect different specific substrates, and whether/how these specific experiences affect the newborn's adaptation/learning depends on the nature of the adaptive/learning task.

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TEACHING PSYCHOLOGY OF LEARNING WITH “SNIFFY” THE VIRTUAL RAT. R. L. Delaney, B. M. Stevenson, & I. B. Johanson, Florida Atlantic University, Boca Raton, FL 33431, USA.

Psychology of Learning is an undergraduate course that is often taught without the benefit of a laboratory component. Since undergraduate psychology/psychobiology students often have a difficult time understanding the principles of learning without the opportunity to train real rats, we decided to incorporate a virtual rat laboratory experience into the class. The “Sniffy Pro” (Wadsworth, 2000) virtual laboratory is a software program designed by Tom Alloway and colleagues to complement courses in learning theory. During the Spring 2000 semester at FAU, we incorporated this virtual laboratory experience into the Psychology of Learning course. Students were given six virtual laboratory homework assignments that counted toward 20% of their overall course grade. As topics such as habituation, sensitization, classical and operant conditioning were discussed in lecture, students were able to simultaneously apply these concepts while training “Sniffy” at home in their own virtual laboratory. Students were required to turn in lab reports which included print-outs of data collected during training (e.g., cumulative records, suppression ratios) as well as a discussion and explanation of their results in the context of the different learning paradigms. In conclusion, our first experience with “Sniffy the Virtual Rat” was an overall success! The majority of students performed very well on their “Sniffy” homework assignments although we noticed an initial difficulty in the computer literacy of some students. In the future, we plan on including at least one computer training session and/or laboratory exercise with “Sniffy” to introduce all students to the software. Overall, 30% of students had favorable comments regarding their experience with “Sniffy” and no students had any negative comments as indicated on their end-of-term course evaluation forms.

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THE EFFECTS OF EARLY STRESS ON ANXIETY DURING ETOH WITHDRAWAL IN ADOLESCENT C3H MICE. J. L. Diaz-Granados, & D. L. Graham, Department of Psychology and Neuroscience, Baylor University, Waco, Texas 76798, USA.

Numerous studies have suggested that there is a relationship between stress vulnerability and alcohol in the adolescent population. In addition, anxiety is a key response observed in animals experiencing ethanol withdrawal. Previous findings from our laboratory indicate that early stress experience increases withdrawal seizure susceptibility in adolescent C3H mice. Therefore, the primary purpose of the present experiment was to investigate the effects of early stress exposure (early weaning) on the anxiety response observed during and immediately after ethanol withdrawal in adolescent C3H mice. In rodents, the normal age of weaning is 22. Early weaned animals have been shown to have significant differences in their stress response and resulting corticosterone levels, as compared to normally weaned animals. In our study, animals were either removed from the mother on PND 16 (abrupt early weaning) or removed from their mother on PND 22. Then, on PND 28 all adolescent animals were exposed to 64 hr of continuous ethanol vapor (via inhalation). Separate groups of animals were tested for their level of anxiety as assessed by their behavior on an elevated plus maze approximately 8 or 24 hr after removal from the chamber. The findings from this study indicate that animals experiencing early stress exhibited a significantly lower anxiety response than those animals normally weaned when tested at the 8 h timepoint ($p < .01$). Conversely, animals experiencing early stress exhibited a higher anxiety response as compared to controls at the 24 h timepoint ($p < .01$). These findings may have important implications with regards to the effect of early stress on the pharmacological response to EtOH in adolescence.

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A TALE OF TWO DIFFERENCES: WORKING MEMORY VS. INHIBITION; DORSOLATERAL VS. VENTROLATERAL PREFRONTAL CORTEX. Adele Diamond, Director, Center for Developmental Cognitive Neuroscience, Eunice Kennedy Shriver Center, & Professor, Departments of Psychiatry & Neurology, University of Massachusetts Medical School.

Results with a task, that is appropriate for children of 4 years through adults, allows us to independently vary demands on working memory and inhibition, and can be used in the fMRI scanner, will be discussed. Differential developmental trajectories for the working memory and inhibitory abilities dependent on dorsolateral prefrontal cortex (DL-PFC) are found. Marked improvements in the inhibitory ability dependent on DL-PFC are found throughout 4–22 years; the ability to hold information in mind appears to be robust early and to show little improvement over that same age range. Evidence will be presented of dissociations among tasks that require DL-PFC but are differentially sensitive to the dopamine content of that region. Not all cognitive tasks dependent on DL-PFC are dependent on dopamine in DL-PFC. Results will be discussed concerning the development of the ability to grasp relations, which appears to depend on ventrolateral PFC, and appears to be impaired in autism. Human infants fail certain tasks, even when there is no delay, because they do not understand the relation between stimulus and reward. Seemingly minor changes in the experimental situation make this relation apparent to infants and enable them to succeed.

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NOVEL-OBJECT-INDUCED CONDITIONED PLACE PREFERENCES IN ADOLESCENT AND ADULT RATS. L. A. Douglas, & L. P. Spear, Center for Developmental Psychobiology & Department of Psychology, Binghamton University, Binghamton, NY 13902-6000, USA.

Novelty exploration can be adaptive in locating biologically relevant stimuli which are important for the survival of the species, such as locating new sources of food, potential mates for propagation, and sources of danger (Bardo, Donohew, Harrington, 1996). Although few animal models of adolescent preference for novelty exist, novelty seeking behavior has been shown in humans to be greater in adolescents than in adults (Arnett, 1994; Gabel et al., 1994; Martin and Robbins, 1995; Trimpopp et al., 1999). In the present study, Sprague–Dawley male and female rats were used in a novel-object-induced conditioned place preference (CPP) to examine developmental differences between adolescents and adults (tested at P37 or P70, respectively) in response to novelty. In addition, the effects of isolate versus grouped housing were also examined across age and gender on place preference and other behaviors. Significant age differences in place conditioning were observed in group-housed animals with adolescents showing novelty CPP, but not adults; in isolated animals, novelty CPP was evident at both ages. Gender also interacted with age to influence conditioning to novelty, with adolescent males (and adult females) showing particularly robust novelty CPP. These preference data are not simply influenced by time spent locomoting on test day. Thus, novelty-seeking behavior in this rodent model is both age- and sex-dependent, as well as sensitive to stress, with isolation stress increasing adult preference behavior to resemble that of adolescents.

Supported by NIAAA grants RO1 AA12150, and R37 AA12525.

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EARLY PATTERNS OF SOCIAL AND MOTOR BEHAVIOR IN BOBWHITE QUAIL. R. F. Eaton, Department of Psychology, The University of Alabama in Huntsville, Huntsville, AL 35899, USA.

Previous studies of bobwhite quail (*Colinus virginianus*) have outlined patterns of perinatal perceptual development in this precocial avian species. The current study examined patterns of social and motor organization during the first 96 hr following hatching. Eight separate groups of chicks, ($n = 9-14$ in each group, total $n = 97$), taken from four different weeks of hatchlings, were socially reared in lighted conditions. The behavior of each group was recorded for 20 min every 8 hr following hatching, from 0 until 96 hr. This procedure yielded a total of 13 sessions of recorded behavior for each of the eight groups. For each group, videotaped recordings were converted to digital files and 10 min ($n = 130$ total min) of each recording were analyzed. Final analyses revealed reliable patterns of huddling and vocalization activity throughout the first 96 hr of postnatal development, as well as increasingly coordinated patterns of social and motor activity. Importantly, changes in huddling and vocalization behavior occurred throughout the first 4 days of life. These behaviors have been previously linked with thermoregulatory and arousal mechanisms in avian and mammalian species. The present results suggest that further investigations of the mechanisms underlying early social and motor behavior in bobwhite quail are needed.

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AFFECT EXPRESSION IN CHILDREN WITH FAILURE-TO-THRIVE. M. J. Finkelstein, R. Biscoglio, S. Werman, E. Lipper, M. J. Ward, & H. J. Polan, Weill Medical College of Cornell University, Ithaca, NY, 10021, USA.

Failure-to-thrive (FTT) is a serious disorder of infancy and early childhood in which children born normal size at term fall below the fifth percentile for weight on National Center for Health Statistics growth charts. Clinical reports and some systematic studies have suggested a link to affective disorder, specifically, higher rates of depression in the mothers of these children and depression-like affect disturbances in the children themselves (Polan, Kaplan, Kessler, et al., 1991; Polan, Leon, Kaplan, et al., 1991). To further investigate this possible link, 211 children between the ages of 9 and 30 months (mean 17.33), 79 with failure-to-thrive and 134 normally growing comparison, were videotaped during play and a mid-day meal with their mothers in the laboratory. The tapes were scored, blinded to age and diagnosis, for affect expression of the children using the Kiddie-Infant Descriptive Instrument for Emotional States (Stern et al., 1992). Principal components analyses yielded three factors, which were interpreted as overall positive affect, feeding problems, and overall negative affect. Mann-Whitney U -tests revealed a significant difference between the groups on Factor 3, negative affect. Unexpectedly, comparison group children displayed more negative affect than FTTs ($p < .05$). A possible model of FTT children as pseudo-compliant with their mothers is discussed.

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DEVELOPMENT AND CONTROL OF MATERNAL RESPONSIVENESS TO RAT PUP ULTRASONIC VOCALIZATIONS. W. J. Farrell, & J. R. Alberts, Department of Psychology, Indiana University, Bloomington, IN, 47405, USA.

There has been more attention paid to the emission of ultrasonic vocalizations (USVs) by infant rodents than to the responses of the dam to these sounds. Our initial experiments compared the behavior of mother rats and virgin females during exposure to a pup that was either warm and silent or cool and vocalizing. The results revealed a potentiated maternal reaction to USVs: mothers approach and maintain proximal orientation to a vocalizing pup far more than do virgin females. Elevated levels of proximal orientation appear within hours of birth, and responsiveness declines by the end of the third week postpartum, around the time of weaning. A regimen of estrogen and progesterone administration facilitates virgin females' proximal orientation toward vocalizing pups. When recorded USVs were used as a stimulus, mothers and virgins responded at similar levels. Mothers did show enhanced proximal orientation toward recorded USVs, however, if a warm, silent pup was positioned below the playback speaker. Pup odors appear to be crucial for maternal responsiveness to USVs. Peripherally induced anosmia attenuated maternal proximal orientation toward vocalizing pups.

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HABITUATION AND CROSS-HABITUATION TO ODORS IN INFANT RATS. M. Fletcher, & D. A. Wilson, Department of Zoology, University of Oklahoma, Norman, OK, USA.

The survival of newborn rat pups depends on their ability to identify their mother almost immediately after birth. In spite of this near-complete dependence on olfaction, there have been no good tests of odor discrimination ability during development. The present study describes a method for testing odor discrimination using habituation and cross-habituation of cardiac orienting responses to odors. Rat pups aged 4, 7, 14, and 18 days old were used. After subcutaneous recording electrodes were implanted bilaterally, the animals were placed in a warmed, humidified chamber and given 20 repeated 4-second odor presentations (mean ITI = 100 s). After the last trial, a different odor was given to test for cross-habituation. The results show that animals as young as PND 4 display bradycardial responses to novel olfactory stimuli, and these responses habituated in all age groups. Finally, all age groups show only partial cross-habituation to the novel odor suggesting some discrimination between odors in this set. Current studies are examining discrimination between more closely related odors, such as homologous series of organic molecules.

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IT TAKES A VILLAGE TO RAISE A COMMUNICATIVE CULTURE: VOCAL TRADITIONS AND COURTSHIP PATTERNS IN COWBIRDS. T. M. Freeberg, A. P. King, & M. J. West, Departments of Psychology and Biology, Indiana University, Bloomington, IN 47405 USA.

Brown-headed cowbirds, *Molothrus ater*, court and mate assortatively by population, and social learning can influence these assortative patterns of courtship. Here, we tested the cultural transmission of vocalizations used in courtship by male cowbirds. Young cowbirds from a South Dakota (SD) population were housed over the winter with adults from the same SD population (SD culture = SDC) or with adults from an Indiana (IN) population (IN culture = INC). Song differences between the original adult SD and IN social models were acquired by the SDC and INC males, respectively, and were transmitted to a second cultural generation of birds. These song differences predicted males' courtship interactions with females during the breeding season, and playback tests to SD females found that they preferred the songs of SDC males to those of INC males. These results connect vocal traditions to courtship decisions and mating success in cowbirds. Furthermore, our studies indicate that complex and dynamic social contexts are needed to determine experimentally how social learning builds communicative cultures in animals.

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EFFECTS OF PRENATAL ETHANOL AND POSTNATAL HANDLING ON CORTICOTROPIN-RELEASING HORMONE AND VASOPRESSIN EXPRESSION. K. I. Gabriel*, M. M. Glavas, L. Ellis, & J. Weinberg, Departments of Anatomy and Psychology,* University of British Columbia, Vancouver, BC, Canada V6T 1Z4.

As adults, animals prenatally exposed to ethanol (E) display hypothalamic-pituitary-adrenal (HPA) hyperresponsiveness (i.e., elevated and/or prolonged plasma hormone responses) to a variety of stressors. One possible mechanism mediating HPA hyperresponsiveness may involve hypersecretion of corticotropin-releasing hormone (CRH) and/or vasopressin (VP) from the paraventricular nucleus (PVN) of the hypothalamus. In contrast, a postnatal manipulation, postnatal handling (H), has been shown to result in lower basal hypothalamic CRH mRNA levels compared to that in nonhandled (NH) animals. The present study investigated the hypothesis that handling would reduce alterations in basal CRH levels in E animals. Sprague-Dawley offspring from prenatal E, pair-fed (PF) and ad lib control (C) groups were assigned to H or NH treatments from 1–15 days of age and terminated under basal conditions at 90–110 days. Brains were collected for CRH and VP mRNA analyses via in situ hybridization. Prenatal ethanol resulted in marginally elevated CRH mRNA levels in the PVN compared to C animals. In addition, handling altered CRH mRNA levels in a sex-specific manner; NH males showed marginally elevated CRH mRNA compared to H males and to H and NH females. There were no effects of prenatal ethanol or handling on VP mRNA levels. These findings suggest that HPA hyperresponsiveness in E rats may be, at least partly, due to altered hypothalamic CRH expression and that postnatal manipulations may be limited in their ability to attenuate such differences. *Supported by grants from NIH/NIAA.*

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ONTOGENETIC CHANGES IN EYEBLINK CONDITIONING AND NEURONAL ACTIVITY IN THE INFERIOR OLIVE. J. H. Freeman Jr., Department of Psychology, The University of Iowa, Iowa City, IA 52242, USA.

The cerebellum and brain stem are essential components of the neural circuitry that mediates eyeblink conditioning in adult organisms (Thompson & Krupa, 1994) and undergo extensive morphological development in the first four postnatal weeks in rats, including the maturation of connections within the olivocerebellar system (Altman, 1972; Crepel et al., 1976). A previous study demonstrated that learning-specific neuronal activity in the cerebellar interpositus nucleus (IN) paralleled the development of eyeblink conditioning (Freeman & Nicholson, 2000). Developmental changes in the neural mechanisms of eyeblink conditioning were further examined by recording the activity of single neurons in the inferior olive during eyeblink conditioning and US-alone trials in infant rats (PND 17 and 24). The percentage of conditioned responses increased as a function of age. The US-elicited activity in the inferior olive on paired CS-US trials and US-alone trials also differed between the two ages. Specifically, there was a greater neuronal response in the 17–18 day-old rats compared to the 24–25-day-old rats throughout training. It is possible that the developmental difference in US-elicited activity reflects the immaturity of inhibitory feedback from the IN to the inferior olive. This interpretation is consistent with the previously reported developmental difference in the recruitment of IN neurons by the US. The development of learning-specific plasticity in the IN may be influenced by the maturation of olivonuclear, nucleo-olivary, and olivo-cortical synaptic efficacy.

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DISTORTING REACTIVATED MEMORIES OF PREVERBAL INFANTS: PASSIVE EXPOSURE EFFECTS. L. Galluccio, A. Wilk, & C. Rovee-Collier, Rutgers, University, Department of Psychology, Piscataway, NJ 08854-8020, USA.

An abundance of research with animals and humans has demonstrated that memory is dynamic and constructive in nature. The memory of preverbal infants is no exception. Passively exposing infants to a novel stimulus immediately after training not only interferes with their newly acquired memory, but also it supplements their training memory so that they subsequently recognize the exposed stimulus. Riccio and Ebner (1981) proposed that reactivated memory is more vulnerable to modification than a newly acquired one. In the sole test of this possibility with infants, however, we had failed to modify 6-month-olds' reactivated memory of the training context using the same manipulation that had so readily modified their newly acquired memory. Recent evidence suggests that the training context may have special significance for infants of this age. Given the importance of memory updating for immature organisms, in the present experiments, we reexamined this question. Three-month-olds were operantly trained for 2 days with a particular mobile, allowed to forget the task, and then were given a reactivation treatment. In Experiment 1A, they were passively exposed to a novel, moving mobile on Day 6, immediately after the reactivation treatment, and were tested 1 day later with either the original mobile (to determine if the exposure impaired the reactivated memory) or the novel one (to determine if the exposure supplemented the reactivated memory). Neither effect was found. The memory performance of the experimental groups and no-exposure control groups was indistinguishable. Because 3-month-olds do not express retention until 24 hr after a reactivation treatment, in Experiment 1B, we asked if reactivated memory had to be active to be modified. Infants were exposed to the novel mobile 24 hr after the Day 6 reactivation treatment and were again tested 1 day later. This time, the novel exposure impaired reactivated memory but did not supplement it. These data reveal that reactivated memory can be modified by passively exposing infants to novel information, but only if the memory is active. Moreover, the impact of passively exposed novel information on a reactivated memory—even when it is active—is less severe than on a newly acquired one. We are currently examining whether actively exposing infants to a novel mobile immediately after a reactivation treatment distorts the reactivated memory.

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MATERNAL CARE AND SEPARATION STRESS IN HIGH- AND LOW-AGGRESSIVE MICE. J.-L. Gariépy, D. Nehrenberg, & R. Mills-Koonce, Center for Developmental Science, University of North Carolina Chapel Hill, NC, USA.

Previous research on the effects of handling in our mouse lines selectively bred for high (NC900) and low (NC100) aggression showed that NC900 males were impervious to such treatment. Adult CORT response in an open field was reduced by handling only in NC100, bringing activation levels down to those measured in the high-aggressive line. The present study examined how neonatal handling affected maternal care and pup USV production in these lines over the first 2 weeks of life. Separation-induced CORT activation and base line levels were also measured in dams and pups on handling days 3, 5, and 7. Daily observations of the litters (12 handled, 12 control) showed that maternal care diminished in both lines over the study period, but was unaffected by handling in NC900. In the absence of handling, NC900 dams licked and manipulated their pups more, and it is only with handling that NC100 dams showed comparable levels. Upon reunion USVs were more frequent in NC100 over the first handling week, and both lines showed a subsequent reduction. CORT activation was initially higher in both pups and dams of the high line as compared to the low line. While measured levels subsequently declined in the high line, concentrations, initially low in the low line, augmented over the same period. The lines did not differ in basal resting CORT levels. Given that selective breeding effects only took place in the low line, these results suggest a Waddingtonian process of genetic "disassimilation" whereby a phenotype ancestrally expressed in advance of environmental stimulation, now requires stimulation in descendant generations for its expression.

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PARADOXICAL EFFECTS OF NEONATAL ALCOHOL AND COCAINE EXPOSURE ON SPATIAL LEARNING. C. R. Goodlett, S. R. Gilmore, & B. S. Neal-Beliveau, Department of Psychology, IUPUI, Indianapolis, IN 46202, USA.

The incidence of infants born with cocaine in their systems has increased greatly in the past few years. Typically, these infants are also exposed to alcohol in utero, since most cocaine users co-abuse alcohol as well. Perinatal exposure to either of these drugs alone has adverse effects on neurobehavioral development, but less is known about the consequences of combined exposure to alcohol and cocaine. The goal of the current study was to determine whether exposure to cocaine and/or alcohol during a critical period of brain development has adverse effects on spatial learning. Long-Evans rat pups were treated with alcohol (3 g/kg), cocaine (10 mg/kg), or alcohol plus cocaine (3 g/kg + 10 mg/kg) once a day from PND1 to PND9. Controls included both an intubated group and an unintubated group. Spatial learning was tested in the Morris water maze on PND26–PND31, comparing place and random location training. Data were collected and analyzed using HVS Image video tracking. Neonatal alcohol treatment significantly impaired acquisition of place learning, whereas neither the combined alcohol/cocaine nor the cocaine group was significantly affected. Probe trial performance did not discriminate the groups. There were no significant treatment effects on random training. The alcohol-induced impairment in spatial learning extended previous findings, showing effects at a lower daily dose than previously studied. Contrary to the predicted exacerbation of the alcohol-induced impairment, combined exposure with cocaine appeared to mitigate the deleterious effects of alcohol on spatial learning. It is unclear why cocaine co-exposure appeared to protect against the deleterious effects of alcohol on development of spatial learning. *Supported by a grant from Faculty Development Office at IUPUI, and NIH grants DA09362 and AA11945.*

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EXPOSURE TO ETHANOL AND NICOTINE DURING THE BRAIN GROWTH SPURT: EFFECTS ON DEVELOPMENT AND SPATIAL DMP PERFORMANCE. T. A. Girard*, H.-C. Xing,† G. R. Ward,† H. Nguyen,† & P. E. Wainwright,† Departments of Psychology* and Health Studies & Gerontology,† University of Waterloo, Waterloo, ON, N2L 3G1, CANADA.

Male Long-Evans rats were reared artificially and, using a 2 × 2 design, were exposed from postnatal days (PND) 6–9 to ethanol (ET; 6.5 g/kg⁻¹/day⁻¹ "binge exposure") and/or nicotine bitartrate (NIC; 6 mg/kg⁻¹/day⁻¹ continuous exposure). Controls were administered maltose-dextrin in amounts isocaloric to ethanol and/or sodium bitartrate. A fifth suckled-control group was reared normally. NIC accelerated eye-opening on PND 14, whereas ET delayed eye-opening and hindlimb support on PND 16. Beginning in postnatal week 7, rats were tested on a spatial delayed matching-to-place (DMP) version of the Morris water maze, which entailed a series of problems, each consisting of search and recall trials, that required the rats to use extra-maze cues to locate a hidden escape platform. In Phase 1 of testing, the ET groups were impaired in the recall trials, but there was no effect of NIC. A longer encoding time (45 vs. 10 s) improved performance across all groups. In contrast, acute administration of nicotine (0.1 mg/kg) immediately prior to testing in Phase 2 failed to improve performance in any group. In conclusion, these results confirm previous findings of impaired spatial DMP task performance in ethanol-exposed rats and further suggest that these memory deficits are amenable to amelioration.

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PRENATAL ALCOHOL EXPOSURE AND POSTWEANING ENRICHMENT ALTER NEOCORTICAL NEURONAL STRUCTURE. J. H. Hannigan, & R. F. Berman, Department OB/GYN, Wayne State University School of Medicine, Detroit, MI 48201 USA.

Environmental enrichment ameliorates behavioral effects of prenatal alcohol by means thought to involve stimulated neuronal plasticity. We reported reduced enrichment-induced hippocampal dendritic spine growth after prenatal alcohol, indicating that alcohol's effects on plasticity endure in hippocampus, despite behavioral amelioration. We hypothesized that other brain area(s) mediate enrichment's ameliorative effects. We examined rat motor area Fr2 because frontal neocortex is implicated in FAS and neurochemical anomalies are reported there after prenatal alcohol in rats and children. Golgi-stained Fr2 layer II/III and V pyramidal neurons were quantified after prenatal alcohol and postweaning enrichment. Prenatal alcohol reduced the effectiveness of enrichment in increasing Fr2 dendritic spine density, suggesting a reduced morphological neuronal plasticity in response to environmental enrichment. The results imply that plasticity elsewhere contributes to improved behavior after enrichment, and/or that other indices of plasticity (e.g., neurotrophin levels and branching) need to be measured. *Supported in part by R01-AA12015, Department of OB/GYN & WSU.*

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REACTIVATION AND REINSTATEMENT SIGNIFICANTLY PROTRACT INFANT MEMORY. K. Hildreth, & C. Rovee-Collier, Rutgers University, Department of Psychology, Piscataway, NJ 08854-8020, USA.

Studies with immature animals have shown that memories can be protracted for long periods of time by reminders. The two types of reminders that have been used are reactivation and reinstatement. A reactivation treatment is a brief exposure to an isolated component of the original training event at the end of the retention interval, whereas reinstatement is an abbreviated practice trial of the original event. Neither alone, without the original experience, produces retention. In the present study with 6-month-old human infants, we compared the effectiveness of reinstatement and reactivation in protracting the subsequent duration of a memory that infants had acquired during operant training. Six-month-olds were trained for 2 consecutive days in either the mobile task or the train task, were allowed to forget the task, and then were given one of the reminder procedures 20 days later. In Experiments 1a and 1b, independent groups of infants received a 2-minute reactivation treatment with either the mobile or train 20 days after training and were tested until they reforgot each task. Infants remembered the reactivated memory of each task for another 2 weeks, forgetting it at the same rate as they had forgotten the original memory. In Experiments 2a and 2b, the preceding procedure was repeated except that infants received a 2-minute reinstatement 20 days after training. In Experiment 2a, infants still remembered the mobile task 4 weeks later, at which point testing was discontinued because they outgrew the task. In Experiment 2b, infants tested with the train from 4 to 11 weeks after reinstatement exhibited significant retention at least 11 weeks after reminding; testing after longer delays is still in progress. These results demonstrate that reinstatement protracts retention at least five times longer than a single reactivation, which had doubled the life of the original memory. Despite recent speculation to the contrary, the difference between these two reminders is not artifactual, and they are not functionally equivalent.

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PRENATAL ETHANOL EXPOSURE ALTERS WET DOG SHAKE RESPONSE TO A 5-HT AGONIST. C. E. Hofmann,^{1,2} M. M. Glavas,¹ K. I. Gabriel,^{1,2,3} & J. Weinberg,^{1,2} Department of Anatomy¹ and Psychology², University of British Columbia, Vancouver, BC, V6T 1Z1, Canada; McLean Hospital, Alcohol & Drug Abuse Research Center, Belmont, MA, 02478, USA.³

Previous research has found that animals prenatally exposed to ethanol (E) show hormonal hyperresponsiveness to stressors. 5-hydroxytryptamine (5-HT) is released in several brain regions in response to stress and has a stimulatory effect on the hypothalamic-pituitary-adrenal (HPA) axis. Data have shown that E animals exhibit 5-HT receptor abnormalities and altered behavior on anxiety and perseveration tests that may be related to 5-HT function. The present study investigated the possibility that 5-HT_{2a} receptor responsivity is altered in E animals. 5-HT_{2a} activation results in wet dog shakes: a paroxysmal shudder of the head, neck and trunk, occurring either spontaneously or in response to certain stimuli, such as administration of the 5-HT₂ agonist DOI [(±)-1-(2,5-dimethoxy-4-iodophenyl)-2-aminopropane hydrochloride]. This behavior may provide a behavioral bioassay for estimating changes in 5-HT_{2a} receptor responsivity. Young female and male Sprague-Dawley rats from ethanol-exposed (E), pair-fed control (PF) and ad lib control (C) dams were tested at 45–60 days of age. Animals were injected with saline or DOI (1 mg/kg SC) at a volume of 1 ml/kg, 2 hr prior to lights off. E females but not males showed a significantly greater rate of DOI-induced WDS compared to PF and C females. There were no differences in WDS in response to saline administration. These results suggest that altered 5-HT receptor function in E females may contribute to altered HPA responsiveness to stressors. Supported in part by an AHFMR award to CEH and AA07789 to JW from NIAAA.

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IS INFANT HANDEDNESS FOR REACHING RELATED TO HANDEDNESS FOR MANIPULATION. T. Hinojosa, & G. F. Michel, Psychology Department, DePaul University, Chicago, IL 60614, USA.

As different manual skills are acquired at different ages during infancy, handedness skills must be assessed separately. Handedness for reaching is the skill most commonly assessed while hand preference for manipulation skills is assessed only occasionally. This study examined the relation between reaching and manipulation handedness by identifying, at 7, 9, and 11 months of age, the manipulation handedness of four groups of 10 infants each. The groups were defined by a valid and reliable infant handedness assessment procedure. One group was consistently right-handed for reaching across the three age periods; another was consistently left-handed; a third showed no preference for reaching; and the fourth group had inconsistent hand-use preferences across the three assessment ages. All infants exhibited no hand-use preference for manipulating objects at 7 months. By 11 months, those with a right preference for reaching showed a right preference for manipulation; those with a left preference for reaching showed a left preference for manipulation whereas those without a reaching preference failed to exhibit a manipulation preference. These results are consistent with the notion that manipulation preferences are influenced in their development by an early manifestation of a reaching preference for the apprehension of objects that is well established by 7 months. Once apprehended, a lack of intermanual object transfer permits one hand to develop manipulation skills with an object thereby expanding the infant's hand-use preference to object manipulation. Handedness is an aspect of hemispheric specialization of function. Thus, these results help reveal how self-generated experiences during infancy can influence the development of some of the processing differences between the hemispheres.

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SOCIAL LEARNING OF ETHANOL CONSUMPTION DOES NOT DEPEND ON LACTATION. P. L. Honey, & B. G. Galef Jr., Department of Psychology, McMaster University, Hamilton, ON, L8S 4K1, Canada.

We conducted the present study to determine if social learning of ethanol preference is dependent upon exposure to ethanol in mothers milk. Rat pups observed their dam, a novel lactating female, or a novel non-lactating female consume either 8% ethanol or water for 8 days (age 18–26 days). Pups were not able to sample the fluids consumed by adults, except through milk of lactating females. As in previous experiments, pups that had observed their dams consuming ethanol demonstrated greater preference for ethanol, after weaning, than did pups whose dams consumed water. Additionally, pups that had observed a novel female consume ethanol also demonstrated an enhanced ethanol preference. There was no difference in the preference levels of pups that were exposed to ethanol through nursing, compared to pups that did not experience ethanol in milk. These results suggest that exposure to an ethanol-consuming adult during weaning is sufficient to induce enhanced ethanol preference, and that exposure to ethanol in milk is not necessary for this preference to be expressed.

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PRENATAL SENSORY EXPERIENCE AND POSTNATAL PERCEPTUAL PREFERENCES: THE PATTERNS THAT CONNECT. H. Honeycutt, & R. Lickliter, Department of Psychology, Virginia Tech, Blacksburg, VA 24061, USA.

Previous studies have indicated that concurrent multimodal stimulation can interfere with prenatal perceptual learning. However, the nature and extent of this interference are not well understood. For instance, rather than characterize this interference-effect as arising from deficits in attentional capacities, it is conceivable that embryos instead attend to features of the multimodal compound as a whole. This study further assessed this issue by exposing three groups of bobwhite quail embryos to (a) no unusual prenatal stimulation, (b) a bobwhite maternal call, or (c) the maternal call paired with light in the period prior to hatching. Experiments differed in terms of the types of stimuli presented during postnatal preference tests (Experiment 1 = familiar call vs. unfamiliar call; Experiment 2 = familiar call + light vs. unfamiliar call; Experiment 3 = familiar call vs. unfamiliar call + light). Embryos receiving no supplemental stimulation showed no preference between stimulus events in all testing conditions. Embryos receiving exposure to a unimodal call preferred the familiar call over the unfamiliar call regardless of the presence or absence of patterned light during testing. Embryos receiving concurrent audio-visual exposure showed no preference between stimulus events in Experiment 1 and Experiment 3, but did prefer the familiar call when it was paired with light during testing (Experiment 2). These findings suggest that concurrent multimodal stimulation does not interfere with prenatal perceptual learning by overwhelming the young organism's limited attentional capacities, but rather biases what information is attended to during exposure and subsequent testing. Furthermore, these results are consistent with the notion that young organisms tend to initially process non-redundant compound stimuli as an integrative unit instead of processing the components of the compound separately.

Research supported by grants RO1-MH62225 and KO2-MH01210 from NIH.

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A RODENT MODEL OF FETAL ALCOHOL-INDUCED DEFICITS IN RESPONSE HABITUATION. P. S. Hunt, & J. S. Phillips, Department of Psychology, College of William & Mary, Williamsburg, VA 23187, USA.

Humans with a history of prenatal exposure to alcohol exhibit a variety of cognitive deficits, including reduced rates of response habituation to novel stimuli. Dysfunctions in basic attentional processes, such as habituation, can have devastating consequences for more complex types of learning, memory, and performance. In this research we employed an animal model of third-trimester human alcohol exposure, administering 5.25 g/kg/day alcohol i.g. to infant rats from PND4–9. Two control groups were either sham-intubated or unhandled during this period. On Day 16 subjects were tested for cardiac orienting and habituation to a novel olfactory stimulus. The alcohol-exposed group was slower to show complete response habituation to the novel stimulus than either control group. In addition, the alcohol pups exhibited longer response latencies, a finding often reported with humans diagnosed with fetal-alcohol syndrome. These results suggest that the postnatal binge model of alcohol administration is effective for producing a rodent model of at least some of the attentional deficits seen in humans exposed to alcohol prenatally.

This research was supported by NIAAA grant AA12143.

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OBSERVATIONS OF MATERNAL BEHAVIOR IN SELECTIVELY BRED MICE. K. E. Hood, & N. A. Dreschel, Departments of Human Development and Biobehavioral Health, The Pennsylvania State University, University Park, PA 16802, USA.

Differences in adult social behaviors in selectively bred lines (Cairns, Gariepy, & Hood, 1990) may be related to line differences in dam-pup interactions. Rodriguez and Gariepy (in progress) indicate that in similar lines, differences in maternal behaviors are observed in a repeated-handling and reunion paradigm. We observed maternal behaviors using undisturbed dams and litters of mice from two selectively bred lines. Pups from the two lines were fostered on the day of birth to dams from a control line (not selectively bred) of the same strain, and dams from the two selectively bred lines were given foster pups from the control line. Observations of 60 litters at two ages (PND 2 and 4) with 20-minute scan samples (12 litters per observation; 5 s per scan) in the light phase of the diurnal sample, and 20 min in the dark phase (20 min after dark onset) were carried out, with high inter-rater reliabilities. In addition, event sampling observations focused on licking by the same dams were conducted in parallel time frames. Preliminary analyses of scan samples suggest that dams from the line selectively bred for high levels of aggressive behavior show more licking of control line pups than do low-aggressive line dams, while control line dams with pups from the high or low aggressive lines do not show differences in maternal behaviors. Event samples show no line differences. Subtle differences in maternal behaviors may be reliably detected in naturalistic observations of undisturbed litters of mice.

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HIPPOCAMPAL FIELD CA1 CELL LOSS FOLLOWING NEONATAL ALCOHOL EXPOSURE IN LONG-EVANS RATS. T. B. Johnson, T. D. Tran, S. Kelley, & C. R. Goodlett, Department of Psychology, IUPUI, Indianapolis, INND 46202-3275, USA.

Binge-like alcohol exposure in rats during postnatal days (PND) 7–9 significantly impairs spatial learning acquisition in the Morris water maze (in juveniles and adults), whereas comparable exposure during PND 4–6 does not (Goodlett & Johnson, 1997; Johnson and Goodlett, 1999). The present study was conducted to test whether place-learning acquisition deficits were associated with loss of hippocampal field CA1 neurons. Neonatal Long-Evans rat pups were randomly assigned within-litter to different treatment groups. Alcohol-exposed pups were given 5.25 g alcohol/kg/day delivered as an 11.9% (v/v) solution in milk formula via two intubations, 2 hr apart, during either PND 4–6 or PND 7–9. Intubated controls were given concurrent intubations with isocalorically matched milk formula throughout PND 4–9, as were alcohol-treated rats on non-treatment days. A third intubation of milk alone was also administered to all groups each day. Mean peak blood alcohol concentrations were approximately 350 mg/dl and did not differ significantly between alcohol treatment groups. Following spatial learning testing (PND 26–31), 45-day-old rats from each neonatal treatment group were randomly selected for analysis of total number of CA1 neurons using the optical fractionator method. From each rat one forebrain hemisphere was embedded in glycol methacrylate. An exhaustive systematic random sample of 30- μ m horizontal sections was saved and stained with cresyl violet for cell counting. Total numbers of hippocampal field CA1 neurons were estimated using a microscope with a 100 $\times$ oil-immersion objective and video camera interfaced with a PC running Castgrid (Olympus, DK) software. Treatment effects on total CA1 cell numbers were observed due to significantly fewer [$F(2,25) = 3.64, p < .05$] CA1 neurons in the PND 7–9 alcohol exposure group, relative to controls (Intubated Control group mean = 2.3×10^5 ; 4–6 group mean = 1.9×10^5 ; 7–9 group mean = 1.67×10^5). These results support the hypothesis that PND 7–9 represents a period of enhanced hippocampal susceptibility to the pathogenic consequences of alcohol exposure. Supported by AA11945.

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A COMPARATIVE STUDY OF MOVEMENT BOUT ORGANIZATION IN PRETERM HUMAN INFANTS, NORWAY RATS, AND SPINY MICE. G. A. Kleven, L. Key, K. Lauer, & S. R. Robinson, Department of Psychology, University of Iowa, Iowa City, IA 52242, USA.

Spontaneous movement, prior to birth, can be observed in the fetuses of all vertebrate species. Although the motor activity of animal and human fetuses subjectively appears random, quantitative analysis has revealed that fetal movements are organized in time and space. Moreover, the temporal and spatial organization evident in fetal motor activity continues to be expressed by animal and human infants after birth. In the present study, we compared the prenatal development of spontaneous limb activity in three species: Norway rats (*Rattus norvegicus*), spiny mice (*Acomys cahirinus*), and preterm human infants. Rats bear altricial young, which are born relatively immature, whereas spiny mice are precocial, their offspring exhibiting mature locomotion within 24 hr after birth. Both rodent species were studied using procedures that involve surgical preparation of the pregnant mother to facilitate direct visualization of fetal behavior. In order to study the potential presence of altricial and precocial aspects of development in preterm infants, the spontaneous movement of 25 human subjects born between 25–28 weeks post-conception, were videotaped at weekly intervals in a NICU environment. All three species showed a significant tendency for limb movements to be organized into discrete multilimb bouts, with the movement of any one limb significantly increasing the probability of movement in another limb. These comparative findings indicate that the development of spontaneous motor activity may exhibit very similar patterns of organization, and by inference, similar underlying mechanisms, in fetal rodents and preterm human infants.

This research was supported by NIH grant HD 33862 and a CASSPR grant from the University of Iowa to SRR.

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NEONATAL CEREBRAL ULTRASOUND FINDINGS AND CONTINUOUS PERFORMANCE TEST RESULTS AT SCHOOL AGE IN VERY LOW BIRTH WEIGHT CHILDREN. J. S. Leach, M. Durkin, & V. J. Hinton, Department of Psychobiological Sciences, New York State Psychiatric Institute, New York City, NY 10032, USA.

The present study examined the predictive value of neonatal cranial ultrasound abnormalities in terms of attention problems at school age in a sample of children born at very low birth weight (VLBW ≤ 1500 g). Sample: 263 children from an ethnically diverse cohort of infants born at VLBW 1991–1993. 26.6% demonstrated at least one type of cranial ultrasound abnormality. Results: *T*-tests were performed comparing mean Continuous Performance Test (CPT) scores. Children who had intraventricular hemorrhage, ventricular megal, or any white matter damage had significantly higher CPT inattention scores than their peers without any positive ultrasound findings. Ventricular megal demonstrated the strongest relationship: $t = -2.458$; $df = 203$; $p \leq .015$. No significant differences in impulsivity scores were found. Discussion: Earlier work with CPT indicates a relationship between high inattention/low impulsivity and ADHD without conduct disorder. Given the lag between birth and the ages at which certain difficulties may be diagnosed it is important to know the relationship between neonatal findings and later outcomes in order to offer information to the parents of VLBW children and to plan interventions proactively.

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MESOPONTINE C-FOS IMMUNOREACTIVITY DURING ACTIVE SLEEP IN WEEK-OLD RATS. J. C. Kreider, & M. S. Blumberg, Department of Psychology, University of Iowa, Iowa City, IA 52242, USA.

Myoclonic twitching of the distal limbs and tail is exhibited by infant and adult rats during a stage of sleep referred to as active sleep in infants and REM sleep in adults. Although some view active and REM sleep as homologous behavioral states, others have argued that they are developmentally distinct. In previous work using decerebrate week-old rats, we showed that active sleep, as measured by myoclonic twitching, was normal when the brain was transected above, but not below, the mesopontine region. This finding corresponds with findings in adults that nuclei within the mesopontine region, such as the laterodorsal tegmental (LDT) and pedunculopontine tegmental (PPT) nuclei, are important for the expression of REM sleep. Therefore, in this study we seek to identify mesopontine structures that are c-fos-immunoreactive during active sleep and wakefulness in week-old rats. Pups are exposed to one of three temperatures for 2 hr: 36°C (thermoneutral), 30°C (moderate cold exposure), or 18°C (extreme cold exposure). It is known that during thermoneutral and moderate cold exposure, pups exhibit high rates of twitching and low rates of awake behavior, whereas during extreme cold exposure pups exhibit low rates of twitching and high rates of awake behavior. The distribution of c-fos expression in the mesopontine region will be determined for each experimental condition. Determining the distribution of fos-positive cells during active sleep will aid in identifying the neural sources of myoclonic twitching and active sleep in infants. Moreover, if LDT and PPT express high levels of c-fos activation in infants during active sleep, then this would provide support for the view that active and REM sleep are homologous behavioral states.

This research was supported by NIMH grant MH50701.

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SPATIAL LEARNING AND MEMORY IN RODENTS: A COMPARISON OF RATS AND MICE OF DIFFERENT AGES. T. J. Levy, & C. R. Almlie, Developmental Neuropsychobiology Laboratory, Washington University School of Medicine, St. Louis, MO 63108-2292, USA.

Historically, rats have been the primary rodent species used for neurobehavioral studies of brain injury and spatial memory. Now, a growing literature elucidates strain and substrain differences in the spatial learning/memory abilities of mice; however, little is known about how spatial memory abilities in mice compare to those of rats. Thus, in Experiment I, two age groups of C57BL6 mice (PND30 and PND50) were tested in the Morris Water Maze (MWM) under the cue (visible platform), place (hidden platform) or random (no contingency) condition. After pretraining, mice demonstrated significant place learning (e.g., shorter escape latencies during acquisition trials) and place memory (e.g., longer probe times during probe trials), compared to mice tested under the random condition. In Experiment II, the spatial learning/memory abilities of mice (Experiment I) were compared to those of rats tested in a MWM starting at PND20, PND50 or PND90. Although some differences were found between mice and rats at different ages for place escape latencies during acquisition trials (potentially due to differential body size and swim speed between species and ages), the only differences between mice and rat age groups for probe trials were longer probe times for PND90 rats versus PND30 mice (significant), and PND50 mice (trend). Further, the overall pattern of performance within each age group of mice and rats was similar, indicating clear spatial learning/memory abilities for both species. Supported by NS35902.

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INCUBATION TEMPERATURE MODULATES POSTHATCHING THERMOREGULATORY BEHAVIOR IN THE MADAGASCAR GROUND GECKO. S. J. Lewis, M. S. Blumberg, & G. Sokoloff, Department of Psychology, University of Iowa, Iowa City, IA 52242, USA.

The temperature at which avian and reptilian embryos incubate influences a variety of developmental and adult characteristics, including gestation length, sex determination, and running speed. Very little is known, however, about the influence of incubation temperature on post-hatching thermoregulatory behavior. We addressed this question by incubating the eggs of the Madagascar ground gecko (*Paroedura pictus*) at temperatures of 24, 26, and 30°C. This species is nocturnal, breeds relatively easily in the lab, and does not exhibit temperature-dependent sex determination. Hatchlings are tested in an apparatus in which they can shuttle between a cold (18°C) and a hot floor (41°C). Within 2 days of hatching, a subject is placed in the apparatus at 6 pm. Beginning at 9 pm, infrared thermographs are acquired every 5 s for 6 hr. When possible, each subject is tested again one and two weeks after hatching. Back-skin temperatures are analyzed from thermographic images that immediately precede shuttling from the cold region to the hot region, or vice versa. Analyses indicate that incubation temperature directly influences these escape temperatures, thus providing strong evidence for epigenetic influences on thermoregulatory behavior in this species.

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PERINATAL VISUAL STIMULATION ALTERS GROWTH, MORTALITY, AND POSTNATAL PREFERENCE IN AFRICAN CLAWED FROGS. K. M. Mann, & M. J. Sleigh, Department of Psychology, George Mason University, Fairfax, VA 22030-4444, USA.

This study examined African clawed frogs' (*Xenopus laevis*) sensitivity to perinatal sensory stimulation. Tadpoles were exposed to patterned visual stimulation from the egg stage until the time of individual testing. At 10 and 15 days of age, control and experimental subjects showed no preference for patterned light versus natural light in a simultaneous choice test. By 20 and 25 days of age, experimental tadpoles preferred patterned light, while control tadpoles preferred natural light ($p < .05$). Although early experience with visual stimulation appears to lead to preferences for it, growth rate data revealed that by 20 days of age, tadpoles exposed to pulsing light were significantly smaller than control tadpoles from the same parents ($p < .05$). In addition, mortality rates were significantly higher for experimental tadpoles than for control tadpoles ($p < .01$), indicating that arousal may be an underlying mechanism mediating the observed differences. These findings lend support to the notion that sensory stimulation that falls within some optimal range maintains normal patterns of development, whereas stimulation beyond the range of the species norm can alter typical development.

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CAUSES AND CONSEQUENCES OF INFANT ABUSE AND NEGLECT IN MONKEYS. D. Maestriperi, The Institute for Mind and Biology, The University of Chicago, Chicago, IL 60637, USA.

Demographic and clinical information obtained from large populations of Cercopithecine monkeys over a period of time of over 30 years has shown that 5–10% of all infants born in a given year are at risk of maternal abuse or neglect. Abuse, but not neglect, is likely to be exhibited by closely related females such as mothers and daughters or sisters, suggesting that it is transmitted across generations along the maternal line. Behavioral observations of infant abuse in rhesus and pigtail macaques indicate that abusive mothers are controlling individuals who may be particularly vulnerable to social stress. Abusive mothers tend to maltreat all of their offspring born in successive years with similar patterns and frequency. Infant cross-fostering studies have shown that abusive mothers maltreat their adopted infants as if they were their biological offspring, suggesting that abusive behavior is a stable maternal trait. Because of some differences in the acoustic structure of the cries of abused and nonabused infants, however, the possibility that some infant characteristics play a role in the occurrence of abuse cannot be ruled out. Abusive mothers are generally interested in infants even more than nonabusive mothers. Moreover, they do not differ from nonabusive mothers in the concentrations of pregnancy and lactation hormones that are known to play a role in maternal responsiveness. Abusive mothers, however, exhibit alterations in stress-sensitive physiological systems that are likely to be the result of adverse early experience. Similar to victims of child maltreatment or individuals with post-traumatic stress disorder, female monkeys who were abused as infants and became abusive mothers themselves exhibit higher levels of CSF concentrations of corticotropin-releasing-hormone (CRH) and serotonin and norepinephrine metabolites, suggesting hyperactivity of extrahypothalamic CRH neurons and dysregulation of the catecholamine systems. Some of these neurochemical variables predict social and maternal behavior in adulthood, including the probability to adopt an unrelated infant in a cross-fostering procedure. The neurobiological alterations associated with infant abuse may play an important role in the occurrence of maladaptive behavior in adulthood, including the perpetuation of infant abuse across generations.

Supported by NIMH grants R01-MH57249 and R01-MH62577.

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EARLY POSTNATAL EXPOSURE TO METHYLPHENIDATE DOES NOT ALTER LATER BEHAVIOR IN THE OPEN FIELD, ELEVATED PLUS MAZE OR WATER MAZE IN CD-1 MICE. M. P. McFadyen,¹ M. R. Penner,¹ N. Carrey,² & R. E. Brown,¹ Departments of Psychology¹ & Psychiatry,² Dalhousie University, Halifax, NS, B3H 4J1, Canada.

Methylphenidate (MPH) is the preferred drug for the treatment of attention deficit hyperactivity disorder in school-aged children. Recent reports indicate that MPH is being prescribed to younger children, sometimes as young as 2 years old. Research to date has not adequately examined the enduring effects of psychostimulant administration during development. Our study examined the effects of exposure during early postnatal development on behavior and learning in adulthood. CD-1 mice were treated daily with MPH (5 or 20 mg/kg), saline or no injections from 3 to 11 days of age. The mice were then tested during adulthood in the open field, elevated plus maze, and Morris water maze. We found that exposure to either dose of MPH had no effect on open field behaviors or on spatial learning in the water maze. Drug-treated animals entered the closed arms of the elevated plus maze less frequently; however, they did not differ from saline or no injection groups on any other measure in this paradigm. These findings suggest that behavioral development, at least in CD-1 mice, is not disrupted by repeated MPH administration in infancy at either of the doses tested. This corresponds well with our previous findings, in which the same mice showed no significant effects of drug treatment on any early measures of neuromotor development when they were tested between 3 and 11 days of age. This research was supported by the NSERC of Canada and by the QEII Hospital research fund.

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FOS EXPRESSION IN THE INFANT PERIAQUEDUCTAL GRAY MORPHINE WITHDRAWAL. A. A. McPhie, & G. A. Barr, Biopsychology Doctoral Program, CUNY Graduate School and New York Psychiatric Institute, New York, NY 10032, USA.

Effective treatments for opiate dependence rely on a comprehensive understanding of the neuronal mechanisms that are involved in opiate dependence and withdrawal. Some of the major brain regions that are involved in morphine withdrawal in the adult rat are the periaqueductal gray, amygdala, nucleus accumbens, locus coeruleus, and the hypothalamus. Each one of these brain regions plays a role in the dependence syndrome, which is defined by unique affective, physical, and physiological signs in both the adult and infant rat. The purpose of the present study was to determine if the PAG is involved in the display of the physical withdrawal signs in the infant rat, through the expression of the nuclear protein Fos. Litters were injected with either morphine sulfate (10 mg/kg) or .9% saline for 6.5 days. On the afternoon of the seventh day they were injected with either the opiate antagonist naltrexone, to precipitate withdrawal, or saline. After 2 hr, each rat was perfused, its brain was then removed and processed for immunocytochemistry. The brains were then sectioned and stained for Fos-like immunoreactivity (Fos-LIR). There were higher numbers of fos-LIR-labeled cells in the PAG, as compared to control rats. The results show that the PAG contains neurons that become active during the appearance of morphine-induced physical withdrawal behaviors, consistent with adult rats, in the 7-day old rat.

Supported by DA-06600 and DA-00325.

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EXPERIENTIAL MODIFICATION OF PHENOTYPES PRODUCED BY SELECTIVE BREEDING. R. Mills-Koonce, & Bauer, Center for Developmental Science, University of North Carolina at Chapel Hill, NC 27599-8115, USA.

Selective breeding for high and low levels of aggression in mice produced two lines which differ markedly in social behaviors. The criterion was the frequency of attacks in a short test involving an isolated subject and a group-reared partner unlikely to lunge the first attack. Males of the high line (NC900) are very reactive and attack unconditionally. The low line (NC100), by contrast, is less reactive and freezes, instead upon social contact. Developmental studies conducted over ontogeny showed (1) that the two lines freeze at an equally high rate early in development but differ in the rate at which freezing declines subsequently, (2) that NC900 is more reactive throughout development than NC100, and (3) that freezing and attacks in the adult constitute a line-specific way of regulating social reactivity following isolation rearing. On this basis, we tested the hypothesis that adults of each line still retain the capacity to either freeze or attack when line-specific modes of reactivity regulation are rendered ineffective. To this end, NC100 males were exposed during a single 2-hour test to the unconditional attacks of NC900 males. NC900 males were first exposed to another male likely to defeat them, and retested on the following day with a different partner. Within 1 hr, 40% of the NC 100 males dominated the encounter, although they never initiated the first attack. NC900 males that were defeated on the first day showed high rates of freezing on the second day and did not attack. These experiments suggested that behavior most sensitive to genetic variation may also be easily amenable to change by variation in experience, a phenomenon that would provide for fail-safe mechanisms in general adaptation

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BEHAVIORAL ACTIVITIES OF PREGNANT SPINY MICE (*ACOMYS CAHIRINUS*) THAT MAY STIMULATE THEIR LATE-TERM FETUSES. J. A. Menge, S. H. Simmonds, & J. R. Alberts, Department of Psychology, Indiana University, Bloomington, IN 47405, USA.

Current conceptions regarding the impact of maternal behavior on fetal development are almost entirely derived from studies of species that give birth to altricial offspring, example, Norway rat (*Rattus norvegicus*). After about a 22-day gestation period, Norway rat offspring are born with relatively immature sensory and motor systems. Studies of the Norway rat have shown that the behavior of pregnant females compresses, vibrates, and accelerates fetuses. Prenatal stimulation has been found to facilitate the acquisition of independent respiration, accelerating onset of suckling, and establishing response patterns to maternal odors. Little is known about the effects of maternal behavior on fetal development in mammals that give birth to precocial offspring, example, spiny mouse (*A. cahirinus*). After about a 38-day gestation period, spiny mice are born with well-developed sensory and motor systems. To investigate the impacts of maternal behavior on fetuses with more advanced sensory and motor development, the activities of pregnant female spiny mice were described. We report our systematic observations of the activities of seven spiny mice (*A. cahirinus*), living with their pair-bonded mate under standardized laboratory conditions. The mice were marked to distinguish sex and their behavior was recorded continuously by time-lapse videography. Frequency and duration of locomotion, grooming, rearing, stirring, scratching, and resting were quantified for the dams 4 and 2 days prior to parturition, as well as on the day of birth. We describe a profile of maternal activity that may provide important stimulation to the perinatal spiny mouse pup.

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STRESS-ELICITED FETAL AND MATERNAL HEART CHANGES IN ANXIOUS VERSUS NON-ANXIOUS WOMEN. C. Monk, W. P. Fifer, R. P. Sloan, M. M. Myers, & L. M. Ellman, Department of Psychiatry, Columbia University, New York, NY 10032, USA.

Recent studies suggest that stress and anxiety during pregnancy affect newborn outcomes, contributing to preterm birth, lower birth weight, and differences in physiology and behavior. However, little research has measured directly the impact of maternal mood on the fetus and characterized the pathways of this association. This study used standardized laboratory procedures that model mildly stressful life events as well as subjective ratings of maternal anxiety to examine the effects of acute maternal stress and sub-syndromal anxiety on fetal heart rate (FHR). Forty-six women in their 36–38th week of pregnancy (mean age 26 years $\pm$ 6) completed the Spielberger State-Trait Anxiety Inventory. Next, continuous ECG, blood pressure (BP), respiration (RSP), and FHR were measured during a quiet rest period followed by a Stroop color-word matching task or an arithmetic task. When FHR responses during maternal stress were compared to the pregnant women's cardiorespiratory reactivity during the same challenge period, a significant inverse relationship between fetal and maternal HR emerged ($r = -.43$; $p < .05$). Furthermore, similar to previous findings from our laboratory, when subjects were dichotomized at the mean into higher than average (ANX(+)) and lower than average (ANX(-)) anxiety groups, FHR changes during maternal stress differed by anxiety groups ($t(44) = 3.15$; $p < .01$). Fetuses of ANX(+) women showed a HR increase while fetuses of ANX(-) women had a HR decrease. These data suggest that (1) fetal and maternal HR changes are correlated during transitory stress. (2) Over the course of gestation, women's stress-elicited cardiac activity may be a pathway by which women's psychological experiences influence fetal development.

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REINSTATING THE NEONATAL SENSITIVE PERIOD FOR OLFACTORY LEARNING. S. Moriceau, R. M. Sullivan, & D. A. Wilson, Zoology Department, University of Oklahoma, Norman, OK 73019, USA.

During a neonatal sensitive period, rat pups must learn their mother's odor in order to form an attachment to their mother. This olfactory learning-based attachment is dependent upon the locus coeruleus (LC) releasing norepinephrine (NE) within the olfactory bulb (OB). Following termination of the sensitive period (10 days old) approach responses become more difficult to condition. It has been hypothesized that the termination of the sensitive period may be due to developmental changes that occur within the LC at about the same time. We report here, in support of this hypothesis that the sensitive period for olfactory learning can be reinstated after the sensitive period has terminated by reinstating the characteristics of the neonatal LC. Specifically, a relative odor preference could be acquired by PND 14 pups by pairing an odor conditioned stimulus with pharmacological manipulations that reinstated the ratio of inhibitory-to-excitatory LC autoreceptor activation in older pups to heighten LC activity. Furthermore, odor associated with OB infusion of an NE β -receptor agonist produced a conditioned approach to that odor even after the end of the normal sensitive period. Together, these results suggest that the end of the sensitive period for attachment olfactory learning may be due to a decrease in stimulus-evoked NE release in the OB from the LC, rather than a change in the OB itself.

Supported by NICHD 33402 and University of Oklahoma Research Funds.

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HOMO IMITANS AND HOMO PROVOCANS: A THEORY OF HUMAN IMPRINTING. E. Nagy, & P. Molnar, Psychiatry and Behavioral Sciences, University of Texas Health Sciences Center, Houston, TX 77030, USA.

Searching for the mechanism of neonatal imitation resulted in the discovery of neonatal initiative capacity, called "provocation." Newborns spontaneously produced previously imitated gestures while waiting for the experimenter's response. Psychophysiological analysis revealed that imitation is accompanied by a heart rate increase while gesture initiation by a heart rate deceleration, suggesting different underlying mechanisms. This result also confirms that muscular activity alone cannot account for the different HR patterns recorded during imitation and provocation. These results also imply that infants are not only capable of "responding" to the interaction but also have the capacity to provoke interactions. These findings may constitute a laboratory demonstration of the processes underlying attachment, and according to our hypothetical model, human imprinting.

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DIFFERENCES IN NUMBER, AMPLITUDE, DURATION, SHAPE AND BOUT STRUCTURE OF SEPARATION-INDUCED INFANT RAT ULTRASONIC VOCALIZATIONS (USV) BEFORE AND AFTER A BRIEF MATERNAL REUNION (POTENTIATION). M. M. Myers, N. Ali, S. A. Brunelli, A. Weller,^a A. Y. Tu, & M. A. Hofer, Department of Developmental Psychobiology, NYS Psychiatric Inst, New York, NY, 10032, USA & ^aDepartment of Psychology, Bar Ilan University, Ramat-Gan, Israel.

Last year we used log-survivor analyses to quantify bout structure of separation-induced infant rat USV. Comparisons between lines selectively bred to emit high vs low rates of USV showed that the major difference between lines was in time between bouts. In this current study USV were analyzed from 28 10-day-old rats from our randomly bred line during 5 min of isolation and again after a 2-minute reunion with their mother. The USV signal was digitized at 100 Hz and analyzed with an interactive graphics program. As expected, USV was "potentiated" following maternal reunion (1st isolation, 351 ± 42 vs. isolation after reunion, 535 ± 49 , $p < .001$). After signal-averaging the USV sound envelope, we found that the amplitude and duration of USV following reunion were greater than during the initial isolation ($+52\%$, $p < .002$; $+16\%$, $p < 0.001$, respectively). Most USV had a sharp peak with symmetric shoulders. However, double peaks, slow rising shoulders, and slow declines were also seen. There was a significant shift in types, with sharp peaks less common after reunion (90 vs. 70%, $p < 0.001$). Results from log-survivor analyses indicated that the average bout duration (i.e., number of USV/bout) was much greater following reunion (2.7 ± 0.4 vs. 11.5 ± 3.1 , $p < 0.02$), but there was no significant change in time between bouts. These results show that maternal potentiation alters many characteristics of USV in addition to number and that maternal reunion alters bout structure in a distinctly different way than selective breeding.

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SELECTIVE BREEDING FOR BEHAVIORAL TRAITS: RELATIONSHIPS BETWEEN STABILIZING AND DIRECTIONAL EFFECTS. D. L. Nehrenberg, J.-L. Gariépy, & D. J. Bauer, Center For Developmental Science, Chapel Hill, NC 27588-8115, USA.

While selective breeding is designed to yield directional effects on target phenotypic traits, little attention has been given to its possible stabilizing influence on the target phenotype and directional effects on correlated traits. This study examines the relationship between directional and stabilizing effects for the expression of attack and freezing behavior (frequencies and latencies) in two lines of mice that have been selectively bred for high and low aggression over the course of 29 generations of selective breeding. Selection based on the expression of a single behavior (level of attack frequency) has produced two different and distinct outcomes on the expression of the target behavior compared to the correlated behaviors, which characterize these two lines of mice. In the high-aggression line (NC900) a directional effect (shift in mean level) was observed only for attack latencies, and a stabilizing effect (attenuation of variance) was observed for attack frequencies. We propose that the evidence for the stabilizing effect on the target behavior (high-attack frequency), in addition to the evidence for decreasing attack latency, supports the occurrence of Schmalhausen's type II selection for attack behavior in NC900 mice. By contrast, directional effects were obtained for both attack and freezing behaviors in the low-aggression (NC100) line. Furthermore, variance for freeze frequency, a behavior that has emerged as a correlated by-product of selecting for low-attack frequency, increased rather than decreased. In addition to the expected directional effect on the attack behavior in NC900 and NC100 mice, we have obtained evidence of a stabilizing effect on attack behavior in the NC900 line and directional effects on the correlated freezing behavior in the NC100 line as a result of selecting for a single behavior.

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RESPONSIVENESS TO PRIMARY TASTANTS AND BEHAVIORAL PLASTICITY IN THE SUCKLING CONTEXT. M. E. Nizhnikov, E. S. Petrov, E. I. Varlinskaya, & N. E. Spear, Center for Developmental Psychobiology, Binghamton University-SUNY Binghamton, NY 13902-6000, USA.

Our recent findings indicate that newborn rat pups increase attachment and ingestive behaviors in response to saccharine available through the surrogate nipple, while reducing it in response to quinine as compared to water (Nizhnikov, Petrov, & Spear, 1999). In the present study we examined whether the pattern of behavior exhibited during the initial exposure to flavored fluids available through the surrogate nipple would be reproduced 1 hr later on a nipple providing water. The data suggest that compared to subjects pre-exposed to the nipple providing water, 1 hr later pups pre-exposed to the nipple providing saccharin increased time attached to the nipple providing water and water intake, while pups pre-exposed to the nipple with quinine, decreased both. Further data indicate that quinine may serve as an unconditioned stimulus (US) in classical olfactory conditioning in the suckling context. Pups decreased responsiveness to the nipple providing water in the presence of the odor (CS), previously paired with intraoral infusions of quinine (US). The data presented suggest that newborn rats can detect primary tastes in the suckling context as early as several hours after birth and accordingly adjust their suckling behavior, while mechanisms of conditioning underlie suckling plasticity.

This research is supported by grants HD36782 to ESP and 5R37MH35219 to NES.

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C-FOS EXPRESSION AFTER ACUTE RITALIN ADMINISTRATION DOES NOT DIFFER ACROSS DIFFERENT STAGES OF DEVELOPMENT IN CD-1 MICE. M. R. Penner,¹ M. P. McFadyen,² R. Pinaud,³ H. A. Robertson,¹ & R. E. Brown,² Departments of Pharmacology,¹ Psychology,² & Anatomy and Neurobiology,³ Dalhousie University, Halifax, NS Canada.

Ritalin (methylphenidate hydrochloride, MPH) is the drug of choice for the treatment of Attention Deficit Hyperactivity Disorder. Previous research has shown that MPH administration affects the adult brain in a manner different from the young brain. In the current study, we set out to determine the target brain regions of acutely administered MPH at different stages of development. On PND 3, 7, 11, 24, and 45, mice were treated with a single injection of saline, 5 or 20 mg/kg of MPH, and sacrificed 1 hr later. Immunohistochemistry for the immediate early gene *c-fos* was performed. Compared to saline-treated controls, mice treated with the high dose of MPH (20 mg/kg) showed dense *c-fos* expression in the striatum and mediodorsal cortices, areas known to be dopaminergic targets. However, in most cases the low dose of MPH (5 mg/kg) produced only weak *c-fos* expression that was nearly indistinguishable from saline treated controls. *C-fos* expression was observed in mice as young as 3 days of age. The results of this study indicate that MPH induces the expression of *c-fos* in the same brain regions as other psychostimulants, including cocaine and amphetamine, and that this expression is especially strong in the dopamine-recipient striatum. However, our results do not support the hypothesis that MPH differentially activates brain regions according to the age of the organism.

Supported by NSERC and MRC of Canada.

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CARDIAC REACTIVITY AND FEAR OF STRANGERS IN POST-INSTITUTIONALIZED CHILDREN. S. W. Parker, J. Bruce, M. Kroupina, & M. R. Gunnar, University of Minnesota; Minneapolis, MN 55455, USA.

Infants reared in institutions typically reach their adoptive families physically stunted and developmentally delayed. Furthermore, a significant number of these children display inappropriate social behaviors, including approaching strangers without hesitation and wandering away from a parent in public without distress, that have been referred to as indiscriminate friendliness. This study sought to examine cardiac reactivity and indiscriminately friendly behavior in children adopted from international orphanages. Furthermore, we examined whether indiscriminate friendliness reflected lack of stranger wariness in these children. Children, adopted into the United States from international orphanages, were assessed at 2 and 8 months post-adoption, and were compared to children matched on age and sex but reared in their families of origin. At both sessions, the child's vagal tone and heart rate were monitored to index the child's emotional response during a stranger approach task. At the second session, parents completed a questionnaire assessing indiscriminate friendliness. The post-institutionalized children did not differ in behavior or cardiac activity in response to the stranger approach task. Like the family-reared children, most post-institutionalized children expressed both negative affect and suppression of vagal tone. More indiscriminate friendliness was described by parents of the post-institutionalized children than family-reared children, however. The findings suggest that indiscriminate friendliness in post-institutionalized children does not result from a lack of wariness of strangers.

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SELF-ADMINISTRATION OF ETHANOL AND SACCHARIN IN NEWBORN RATS: EFFECTS ON SUCKLING PLASTICITY. E. S. Petrov, E. I. Varlinskaya, & N. E. Spear, Center for Developmental Psychobiology, Binghamton University – SUNY, Binghamton, NY, USA.

Our previous studies have shown that cesarean-delivered rats prior to any suckling experience voluntarily ingest ethanol and saccharin from the surrogate nipple. The present study investigated peculiarities of attachment and ingestive behaviors toward the surrogate nipple providing water, ethanol, or saccharin during each of the first three postnatal days (PND 0–2). The results of the experiments demonstrated that not only cesarean-delivered neonates tested on PND0 prior to any suckling experience but also vaginally delivered newborns enriched with regular suckling experience and tested on PND1–PND2 attached to and ingested fluids from the surrogate nipple. Neonatal suckling behavior differed as a function of fluid available from the nipple, verifying that neonatal rats differentiate water, saccharin, and ethanol. Paired experience with the surrogate nipple providing saccharin or ethanol (but not experience with the fluids alone, the nipple alone or nipple providing water) resulted in relatively permanent enhancement of responsiveness to the nipple itself, observable even 24 hr after pre-exposure. Experimental data provide strong evidence that in newborn rats an encounter with ethanol in the context of suckling behavior represents a significant incident and suggest strong appetitive reinforcing effects of ethanol early in life. The research was supported by grants AA11960 and AA12384 to NES and HD76782 to ESP.

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COMPETENCE OF NIPPLE ATTACHMENT AT BIRTH: IMPORTANCE OF DOING IT THE HARD WAY. H. J. Polan, L. Eljuga, J. Lee, S. Thomas, & M. A. Hofer, New York State Psychiatric Institute, New York, NY 10032, USA.

Even before the first nursing bout, we found that newborn rats perform competent maternally directed orienting behaviors (MDOBs) that result in ventral-ventral contact with their dam, when she is encountered in a naturalistic position above them (Polan, Milano, Eljuga, & Hofer, 2000). These results led us to reexamine earlier findings that oral nipple attachment is performed poorly at birth, acquiring competence over the next 12–24 hr (Dollinger, Holloway, & Denenberg, 1978). In that work, hand-held pups were tested for nipple attachment to a supine dam, possibly distorting their normal orienting cues. To mimic the geometry of the pup's encounter with its dam in the nest as she hovers over the litter to nurse, we placed pups prone inside a corral and lowered the anesthetized dam from above till her ventrum lightly contacted the pup's dorsum. Pups tested singly for 4 min had access to three nipples within the corral, and their latency to orally grasp was recorded. Any nipple grasped was shifted out of the corral and replaced by a fresh one for the next pup. Three pups from 20 litters were tested; five litters at each of four ages: (1) on the day of birth (P0) before the first nursing bout, (2) on the day of birth after the first nursing bout, (3) on P1, or (4) P2. Half of pups at each age attached to a nipple during the test, with a mean latency of 103 (*SD* 65) s, with no differences across age groups. In this naturalistic model, nipple attachment is seen to be competent at birth, assisted by the performance of competent MDOBs.

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EFFECTS OF PRENATAL SENSORY STIMULATION ON BEHAVIORAL AROUSAL IN BOBWHITE QUAIL EMBRYOS. G. D. Reynolds, & R. Lickliter, Department of Psychology, Virginia Tech, Blacksburg, VA 24061, USA.

Although a number of studies have demonstrated the effects of prenatal experience on subsequent behavioral development, how these effects are achieved remains a topic of enduring interest. The present study examined the immediate effects of unimodal and bimodal prenatal sensory stimulation on behavioral arousal in bobwhite quail embryos. The top portion of the eggshell was removed on the 21st day of incubation and embryos were videotaped during a 12-minute exposure period to (a) unimodal auditory stimulation, (b) unimodal visual stimulation, (c) concurrent audio/visual stimulation, or (d) no supplemental stimulation. Recordings of a bobwhite maternal call served as the auditory stimulus and patterned light transmitted via a strobe light served as the visual stimulus. Videotapes were analyzed to determine the amount of time embryos were active and the average duration of activity during stimulation and nonstimulation conditions. Results indicated that embryos' overall activity levels can be significantly affected by the type of prenatal sensory stimulation provided. Embryos receiving no supplemental stimulation or unimodal auditory stimulation showed relatively low levels of activity, whereas embryos receiving unimodal visual or bimodal audio/visual stimulation exhibited significantly higher levels of activity. These findings are consistent with the neuroembryological development of the sensory systems, in that the auditory system is functional prenatally, whereas the visual system is not typically functional until after hatching. Bobwhite embryos appear to be particularly sensitive (in real time) to unusually early visual stimulation and to concurrent bimodal stimulation, paralleling previous studies from our lab reporting that altered prenatal sensory stimulation can affect subsequent postnatal (developmental time) perceptual and behavioral responsiveness.

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IDENTIFYING THE NEURAL CIRCUITS MEDIATING ABNORMAL REPETITIVE BEHAVIOR. M. F. Presti, & M. H. Lewis, Departments of Psychiatry, Psychology, and Neuroscience, University of Florida, Gainesville, FL 32610, USA.

Abnormal repetitive behaviors, or stereotypies, are often observed in conjunction with developmental disorders such as autism and mental retardation. It has been suggested that these behaviors are expressed as a result of inappropriate facilitation of motor programs initiated by the primary motor cortex. The present work employed an animal model of developmental disorders to attempt to identify the neural circuits mediating stereotypy. The mice used are ideally suited to study the development of stereotypy, because their behavioral phenotype is sensitive to housing conditions. The majority of mice reared in standard laboratory cages develop spontaneously stereotypic behavioral repertoires, while most mice reared in an enriched environment do not exhibit stereotypy. It is thought that stereotypy resulting from environmental restriction may be expressed due to alterations in the excitability of motor circuits traveling through the basal ganglia. The present study evaluated this hypothesis by attempting to attenuate spontaneously emitted stereotypy by inhibiting the excitatory corticostriatal efferents which initiate activity along the implicated motor circuit. Bilateral striatal infusion of the non-competitive NMDAR antagonist MK-801 produced a dramatic reduction in stereotypy levels, but did not inhibit normal locomotor behavior. These findings support the hypothesis that stereotypy arises due to elevated activity along motor circuits of the basal ganglia. It is postulated that environmental restriction results in increased excitability along these circuits, which facilitates the maintenance of initiated motor programs, resulting in stereotypy.

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HYSTERESIS IN A MODEL OF INFANT FREE LOOKING. S. Robertson, J. Guckenheimer, A. Masnick, & L. Bacher, Department Human Development, Cornell University, Ithaca, NY 14853, USA.

Visual exploration of the environment, which entails repeated shifts of gaze and attention, is an activity of singular perceptual-cognitive importance for the young human infant, yet very little is known about its underlying dynamics. In Study 1, we built a dynamical model to simulate the free looking behavior of 1-month-olds. We made minimal assumptions: bistability (looking-looking away) with noise and bias. The model was able to reproduce the rate at which infants shift between looking at and away from static objects and, at the same time, the median duration of looking. But the model could not also reproduce the median duration of looking away. Analysis of the model's behavior revealed an excess, compared to infants, of very rapid transitions between looking and looking away. In Study 2, we added hysteresis to the model which made the boundary between looking and looking away depend on the direction of the transition. With the addition of hysteresis the model was able to simultaneously reproduce seven key measures of infant free looking: (1) the transition rate between looking and looking away, (2) the median durations of looks and looks away, and (3) the proportions of looks and looks away less than 1 s and longer than 10 s. The results suggest that hysteresis may be an important property of shifts of gaze or attention in young infants, which could help explain normal characteristics of early behavior (e.g., obligatory attention) and predict later dysfunction (e.g., aspects of attention deficit disorders).

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MATERNAL BEHAVIOR IN RATS SELECTIVELY BRED FOR INFANT ULTRASONIC VOCALIZATION (USV). H. Rojowsky, A. Weller, M. A. Hofer, & S. A. Brunelli, Department of Developmental Psychobiology, NYS Psychiatric Institute, New York, NY 10032, USA.

This study explored differences in maternal behaviors in rat mothers selectively bred for high or low rates of isolation-induced USV in infancy. Maternal behavior of dams in the 11th–13th generations of breeding were observed under undisturbed conditions on postnatal days 4, 5, and 10. No differences were found between lines at PND 5. Undisturbed high and low line dams spent more time in the nest and more time nursing pups than random line mothers. The proportion of nest time spent nursing did not differ among the lines. Neither the rate nor total number of milk ejections differed significantly, either. Mothers were also observed on PND 10 after disturbance due to routine screening of litters for USV. Prior to pup screening, dams were removed from the home cage, and individual pups were isolated for 2 min in a novel environment. Pup weights and temperatures did not differ between lines. After testing, pups were scattered in the home cage and mothers were returned to the home cage for 32 min observation. Pup USV in the home cage was recorded during this time. Pups from the three lines did not differ in USV emitted in the home cage. High line dams retrieved significantly fewer pups than random or low line dams, and significantly more high line pups returned to the nest on their own. These line differences in maternal behavior may be a function of maternal phenotype, or of pup phenotype, or an interaction of both.

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ENDOGENOUS OPIOIDS ARE NECESSARY FOR AN ODOR PREFERENCE DURING AVERSIVE MOTHER-INFANT INTERACTIONS. T. L. Roth, & R. M. Sullivan, Department of Zoology, University of Oklahoma, Norman, OK 73019, USA.

Rat pups learn maternal odor through associative learning during a temporally defined sensitive period (ends at postnatal day (PND) 10). This learned odor preference mediates behaviors important to survival. Failure to learn the maternal odor preference results in a greatly diminished survival rate. *Paradoxically, odor-moderate shock pairings also result in an odor preference during the neonatal sensitive period, but an odor aversion after PND10.* The present experiment assessed the neurochemical basis of the odor preference induced by paired presentations of odor-moderate shock. Recent research has indicated that noradrenergic input is not necessary for learning an odor preference during odor-shock conditioning, suggesting the importance of another learning system. Additionally, studies have demonstrated the importance of opioids in early olfactory learning, but have not examined their role using a naturalistic model of mother-infant interaction. In this study, PND8 and PND12 pups were systemically injected with naltrexone, a non-specific opioid antagonist, or saline and randomly assigned to receive either paired or backward presentations of odor-moderate shock, or odor-only presentations. Pups were then tested for an odor preference or aversion using a Y-maze. Results indicated that blocking the opioid system during odor-shock conditioning in PND8 pups disrupted learning an odor preference, but did not disrupt learning an odor aversion in PND12 pups. Control subjects showed neither an odor preference nor aversion. These results suggest that the opiate system is necessary for early olfactory learning during natural mother-infant interactions and preference formation to noxious stimuli during the sensitive period. Supported by NIHHD 33402 and University of Oklahoma Research Funds.

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EFFECTS OF ADRENERGIC BLOCKADE ON POSTPARTUM ADAPTIVE RESPONSES INDUCED BY LABOR CONTRACTIONS. A. E. Ronca,¹ N. A. Mills,² K. P. Lam,² & L. E. Hays,³ ¹Life Sciences Division, NASA Ames Research Center, Moffett Field, CA 94035, USA; ²DeAnza College, Cupertino, CA 95014, USA; ³Department of Biology, University of Massachusetts, Amherst, MA 01002, USA.

Prenatal exposure to labor contractions augments the expression of postnatal adaptive responses in newborn rats (Ronca, Abel and Alberts, 1996). Near-term rat fetuses exposed prenatally to simulated labor contractions and delivered by cesarean section breathe and attach to nipples at greater frequencies than non-stimulated fetuses. Plasma NE and EPI was significantly elevated in newborn rats exposed to vaginal birth or simulated labor contractions (compressions) with cesarean delivery as compared to non-compressed fetuses (Ronca, Abel, Renner, Ronan, & Alberts, 1999). In the present study, we investigated adrenergic mechanisms underlying labor-induced postnatal adaptive responses. Following spinal transection of late pregnant rat dams, fetuses were administered neurogenic or non-neurogenic adrenergic blockade: (1) bretylium (10 mg/kg sc) to prevent sympathetic neuronal release, (2) hexamethonium (30 mg/kg) to produce ganglionic blockade, (3) phenoxybenzamine (10 mg/kg sc), an α -adrenergic receptor antagonist, (4) ICI-118551 (10 mg/kg sc), a β receptor antagonist, or (5) vehicle alone. Fetuses were either compressed (C) or non-compressed (NC) prior to cesarean delivery. α - and β -adrenergic antagonists reduced respiration and nipple attachment rates, while sympathetic antagonists and vehicle alone did not. These results provide additional support for the hypothesis that adaptive neonatal effects of labor contractions are mediated by adrenal and extra-adrenal catecholamines. Supported by MH46485.

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DEFERRING IMITATION PROTRACTS RETENTION BY 6-MONTH-OLD HUMAN INFANTS. C. Rovee-Collier, J. Heravi, & R. Barr, Rutgers University, Department of Psychology, Piscataway, NJ 08854-8020, USA.

The time window is a period within which additional information can be integrated into an existing memory. Prior research has shown that the later information is presented within the time window, the longer it protracts retention. Three-month-olds typically remember a mobile task for 5 days. When infants were given an additional 3 min of training (a reinstatement) immediately after training their retention was extended by 1 day (to 6 days). However, when infants were given a reinstatement at the end of the time window, 5 days after training, their retention was extended an additional 16 days (to 21 days). Using deferred imitation tasks, immediate imitation of the target actions does not protract retention after test delays ranging from 10 min to 4 months. The question for the present experiment was whether deferring imitation of the target actions until the end of the time window might protract retention for a puppet imitation task that infants typically remember for 1 day. In the present experiment, an adult modeled three target actions on a puppet, and 6-month-old infants imitated those actions three times either immediately (*immediate imitation group*) or 24 hr later (*deferred imitation group*). The *immediate imitation group* performed significantly above a baseline control group when tested 1 but not 2 or 3 days after modeling. The *deferred imitation group*, however, now exhibited retention of the target actions 3 days after modeling. Moreover, deferring imitation until the end of the time window may protract retention even further. We are currently testing another deferred imitation group 5 days after modeling and will continue to test until they have forgotten. These findings reveal that deferred imitation may play a functional role in protracting retention.

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BEHAVIORAL CONSEQUENCES OF NEAR-TOTAL NEONATAL STRIATAL DA DENERVATION: COMPARISON OF P0 VS. P7 6-OHDA LESIONS. R. N. Sahr,^{1*} & B. S. Neal-Beliveau,^{1,2}

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It has been suggested that both timing and degree of striatal dopamine (DA) depletion are critical factors in determining the long-term consequences of neonatal 6-hydroxydopamine (6-OHDA) lesions. In earlier studies, adult animals depleted of >70% striatal DA on day of birth (PND0) exhibited hyperactivity following treatment with the D1-like receptor agonist SKF 38393 (Neal-Beliveau and Joyce, 1998), while animals depleted of >70% DA on PND7 exhibited increases in grooming, but no hyperactivity (Neal-Beliveau and Joyce, 1999). Our goal was to compare the consequences of near-total PND0 vs. PND7 lesions ($\geq 90\%$ loss of TH immunostaining). Rats received bilateral intra-striatal (i.s.) injections of 6-OHDA (20 μ g per side) on PND0 or PND7. At PND40, rats were tested for baseline locomotor activity and stereotypical behaviors (grooming and licking), followed by three every-other-day injections with SKF 38393 (10 mg/kg). PND0 and PND7 rats demonstrated equivalent hyperlocomotion compared to controls after SKF 38393. Also, both lesioned groups displayed comparable levels of stereotypical licking, which was not observed in controls. Stereotypical grooming was not significantly increased in either lesioned group, and was only elevated in animals whose degree of DA depletion was <90%. These results indicate that near-total DA denervation, unlike partial denervation, does not produce differential behavioral consequences in PND0 versus PND7 lesioned rats. Analysis of D1 binding with quantitative receptor autoradiography is currently underway to compare the effects of each lesion day at the receptor level. Supported by MH51413.

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NMDA & GABA_A RECEPTORS: THEIR ROLE IN THE RESPONSIVITY AND TOLERANCE TO ETHANOL DURING ONTOGENY. M. M. Silveri, & L. P. Spear, Department of Psychology and Center for Developmental Psychobiology, Binghamton University Binghamton, NY 13902-6000, USA.

The role of NMDA and GABA_A receptor systems in the responsivity and tolerance to the hypnotic and hypothermic effects of ethanol (EtOH) during ontogeny were examined. Animals at postnatal day (PND)26 or PND70, were given an intraperitoneal (ip) injection of an NMDA receptor antagonist (+MK-801), GABA_A receptor agonist (muscimol), or saline 10 min prior to ip injection of EtOH (3.5 or 4.5 g/kg) or isovolumetric saline and sleep time (ST: time interval between loss and regain of the righting reflex) and waking tail blood alcohol levels (BALs) measured. On Day 2, all animals were challenged with EtOH, and LORR and ST, waking BALs and BrALs determined. Sensitivity to EtOH increased with age, as observed previously (Silveri & Spear, 1998, 1999; Little et al., 1997). +MK-801 potentiated EtOH sedation similarly across age, whereas immature animals were more sensitive to muscimol's potentiating effect on ST than mature animals. Consistent with previous studies, only PND26 animals displayed acute tolerance (AT) to EtOH on Day 1 (indexed as a positive slope of the linear regression of waking BALs on ST). This expression of AT at PND26 was blocked by +MK-801, but not muscimol. Thus, increasing EtOH sensitivity to adult levels using muscimol at PND26 did not influence AT, suggesting that AT develops independently of EtOH sensitivity. Conversely, blocking AT using +MK-801 did not attenuate age differences in ST, suggesting that ontogenetic patterns of EtOH response are not simply a function of the amount of AT present. Regardless of pharmacological treatment, rapid tolerance was observed at PND70 but not PND26; when combined with the AT data these findings are evidence for a dissociation in the development of acute and rapid tolerance during ontogeny. Supported by NIAAA grant R37 AA12525-01.

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ADULT BEHAVIORAL TESTS OF RATS SELECTIVELY BRED FOR INFANTILE ULTRASONIC VOCALIZATION. H. N. Shair, S. A. Brunelli, Z. Velazquez, & M. A. Hofer, Department of Psychiatry, Columbia University and Division of Developmental Psychobiology, New York State Psychiatric Institute, New York, NY 10032, USA.

The effects on adult behavior of selective breeding for infantile USV were examined in a series of experiments. Three lines of rats have been selectively bred for 17 generations for the rate of isolation-induced ultrasonic vocalizations (USV) at postnatal day (PND) 10: high and low vocalizing animals (HI, LO) and their randomly bred controls (RA). This research is the first breeding program for a mammalian infantile trait. Thus, it was not known whether there would be extensive effects throughout the lifespan, or what the pattern of any effects might be. Several aspects of adult behavior were tested. In a test of *adult vocal behavior*, tactile stimulation induced USV in 50% of the animals. In those vocalizing, there was a trend for HI rats to call at a greater rate than LO rats. One adult test we have used in previous generations, the *elevated plus maze*, has been validated as an anxiogenic environment. In this generation, HI rats entered the open arm more often than RA. During the Porsolt swim test, a measure of depressive-like behavior, HI rats spent more time immobile than LO line animals. Finally, while there were no differences in blood levels of corticosterone immediately after 30 min of *restraint*, heart rates of restrained rats did vary by line. HI and LO rats were higher than RA during the initial epoch. After that, the heart rates of HI and RA gradually fell, but LO animals maintained tachycardia throughout restraint. These data suggest that breeding for infantile USV affected relatively specific aspects of adult behavior and physiology.

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AMOUNT OF PRENATAL VISUAL STIMULATION ALTERS HATCHING RATE AND POSTNATAL BEHAVIOR IN LEOPARD GECKOS

(*EUBLEPHARIS MACULARIUS*). M. J. Sleight, Department of Psychology, George Mason University, Fairfax, VA 22030, USA.

This study examined leopard geckos' sensitivity to augmented amounts of prenatal sensory stimulation. Embryos were exposed to patterned visual stimulation beginning at either 1 or 2 weeks prior to hatching. Embryos exposed to the substantially augmented amount of prenatal visual stimulation hatched significantly earlier than embryos which were provided either the moderately augmented or no additional amount of prenatal visual stimulation ($p < .01$). Hatchling mass was not affected. In addition, embryos exhibited a postnatal preference for the stimuli experienced prenatally. Embryos exposed to substantially augmented visual stimulation demonstrated a postnatal preference for patterned light in a simultaneous choice test at 24 hr of age ($p < .01$). In contrast, control embryos demonstrated a preference for normal light conditions ($p < .01$), while experimental embryos exposed to lesser amounts of prenatal visual stimulation did not show a preference for either stimulus. At 1 week of age, geckos in all conditions failed to exhibit a preference for either stimulus, suggesting that the previously observed preference was short-lived. These findings lend support to the notion that sensory stimulation that falls within some optimal range maintains normal patterns of development, whereas stimulation beyond the range of the species norm can alter species-typical patterns of development.

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SPONTANEOUS EYEBLINKS IN HUMAN INFANTS AS AN INDEX OF DOPAMINE ACTIVITY. W. Smotherman, L. Bacher, N. Korba, & L. Tellefsen, Department of Psychology, Binghamton University, Binghamton, NY, USA.

Dopamine (DA) activity is positively related to changes in the rate of spontaneous eyeblinking (SB) in non-human primates, and clinical studies support DA's role in SB. In human adults, the rate of SB is sensitive to arousal and many visuomotor tasks. Little is known about the rate of SB in human infants other than that the rate is much lower in infants than adults. Our objective was to study the rate of SB in human infants during different tasks. Data from 10–12-week-old infants from two different studies were analyzed. In a study of feeding, 11 infants were observed before, during, and after bottle feeding. In a study of attention, 10 infants were observed during ad libitum looking at static visual stimuli; nine infants were observed during looking before and after new stimuli were presented. The SB rate was greater during bottle feeding than before feeding ($p = .05$) and after feeding ($p = .01$). The SB rate was greater after the new visual stimuli were presented ($p = .02$), but not after no new stimuli ($p = .78$). These results are consistent with new theoretical views on the functions of DA and also suggest that SB might be an easily obtained measure of DA activity in human infants. Research supported by MERIT Award HD 16102 to WPS.

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EFFECTS OF LOCUS COERULEUS STIMULATION ON OLFACTORY CONDITIONING IN THE NEONATAL RAT. G. Stackenwalt, D. A. Wilson, & R. M. Sullivan, Zoology Department University of Oklahoma, Norman, OK 73019, USA.

Neonatal olfactory learning is critical for early mother–infant attachment and is accomplished through associative learning. Specifically, pairing maternal or novel odor with maternal stimuli (e.g., tactile stimulation, milk or warmth) is sufficient to result in a preference for that odor. Previous research from our lab has shown the importance of norepinephrine (NE) in this conditioning. For example, lesioning the neonatal LC prevents odor preference acquisition. The purpose of the present experiments is to strengthen further the link between NE and learning by manipulating the source and target of NE within the brain of PND rat pups. In Experiment 1 bilateral cannulas were implanted in the locus coeruleus (LC) and infused with acetylcholine (2 mM ACh) which increases LC firing and subsequent NE release. In Experiment 2 bilateral cannulas were implanted into the olfactory bulb (OB) and infused with NE-receptor agonist, isoproterenol (50 mM). In each experiment, a citral olfactory conditioned stimulus (CS) accompanied the LC and OB NE activation. The next day, pups were tested for an odor preference using a 2-odor choice test (CS vs. familiar odor). The results indicate that activation of the LC or direct infusion of NE agonist during an odor presentation results in a subsequent preference for the odor. The results further suggest that NE from the LC is capable of supporting learning in infant rats. Supported by NICHD 33402 and University of Oklahoma.

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SODIUM NITROPRUSSIDE DECREASES BLOOD PRESSURE AND EVOKES ULTRASOUND PRODUCTION IN INFANT RATS. G. Sokoloff, M. S. Blumberg, T. A. Gorby, S. J. Lewis, & R. F. Kirby, Department of Psychology, University of Iowa, Iowa City, IA 52242, USA.

Infant rats emit ultrasonic vocalizations (USVs) when isolated from the nest. Under controlled conditions, USVs are emitted during extreme cold exposure and after administration of clonidine, and we have hypothesized that both of these methods evoke USVs by decreasing venous return. Previous work has shown that clonidine evokes maximal USVs in 15-day-old rats, an age that also corresponds with maximal sympathetic tone. To further examine the effect of cardiovascular manipulations on the production of USVs at different ages, we administered sodium nitroprusside (SNP; 400 μ g/kg), a direct-acting dilator of arteries and veins, to 8-, 15-, and 20-day-old rats. SNP elicited a rapid and brief baroreflex-mediated tachycardia at all ages tested; the magnitude of the tachycardia increased with age. Only 15-day-olds emitted significant rates of ultrasound production, and the period of vocalization corresponded with the cardiovascular changes induced by SNP. To further assess the effects of SNP on the cardiovascular system, we next measured arterial pressure in unanesthetized 8-, 15-, and 20-day-old rats after administration of SNP. For all ages, SNP resulted in a rapid and brief decrease in blood pressure. For 8-day-olds the change in blood pressure was small, consistent with the magnitude of the tachycardia. 15- and 20-day-olds showed larger decreases in blood pressure, although the magnitude of the tachycardia was greater in the older infants. These results suggest that 15-day-old rats may be maximally sensitive to cardiovascular challenge as not only clonidine but also SNP result in more USVs at this age. This research was supported by NICHD grant HD38708.

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EFFECT OF PRIME DURATION ON A REACTIVATED MEMORY IN HUMAN INFANTS. B. Sweeney, A. Joh, & C. Rovee-Collier, Rutgers University, Department of Psychology, Piscataway, NJ 08854-8020, USA.

In the present research, we asked what was the minimal duration of a reactivation treatment that would alleviate forgetting for 3- and 6-month-old infants. Previously, we used 180- and 120-second reactivation treatments to alleviate forgetting in 3- and 6-month-old human infants, respectively. However, these exposure durations were arbitrarily chosen; briefer primes might also be effective. For nonhuman primates, evidence of visual recognition memory (significantly longer looking at a novel than at a familiar stimulus) requires a 5-second familiarization period. For 5–6-month-old humans, the minimal familiarization ranges from 5 to >20 s, depending on the complexity of the stimulus. These data suggest that a considerably shorter prime might alleviate forgetting. In Experiment 1, 6-month-olds were trained to move a mobile by kicking, were allowed to forget the task, and were given a reactivation treatment with the mobile 20 days later. The reactivation treatment (priming) lasted 30, 15, 7.5, 5, or 3.5 s. Infants primed for 7.5 s or longer exhibited significant retention 24 hr later, but infants primed for less than 7.5 s exhibited none. The 7.5-second prime was also effective 24 hr later for 3-month-olds. Two succeeding experiments examined whether reactivation is an all-or-none process. If so, then any duration of an effective reactivation treatment should affect subsequent retention equivalently. In Experiment 2, we asked whether a 7.5-second prime recovers the forgotten memory at 6 months as rapidly as a 120-second prime. After a 120-second prime, infants exhibit retention within 1 hr whether trained in the operant mobile or operant train task. After a 7.5-second reactivation treatment, however, infants exhibited no retention 1 hr later. In Experiment 3, we asked if the recovered memory is as persistent after the 7.5-second prime as after a 120-second prime. Whereas 6-month-olds exhibit retention for 13 days after a 120-second prime in both the mobile and train tasks, they exhibited no retention 13 days later when given the 7.5-second prime. Taken together, Experiments 2 and 3 reveal that reactivation is not all-or-none: A briefer (i.e., weaker) exposure to a reminder takes longer to recover a forgotten memory and sustains the reactivated memory for a shorter period. We are currently attempting to pinpoint how fast the 7.5-second prime recovers forgotten memory and how long it sustains reactivated memory.

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TESTOSTERONE TRANSIENTLY INCREASES DOPAMINE RECEPTOR DENSITY IN MALES AND FEMALES DURING THE PERIADOLESCENT PERIOD. M. H. Teicher, A. P. Thompson, & S. L. Andersen, McLean Hospital & Harvard Medical School, Belmont, MA 02478, USA.

Lifespan changes in symptoms and their severity of Attention Deficit/Hyperactivity Disorder (ADHD) and Tourettes Syndrome coincide with hormonal changes. Symptoms for both disorders get worse during puberty, but then wane during young adulthood. During the periadolescent period, dopamine receptors are over-produced and eliminated during transitions through childhood and adulthood. We have previously hypothesized that this process may serve as a permissive factor for the expression of these disorders. Furthermore, sex differences in this process, where males over-produce striatal dopamine receptors while females do not, offer an explanation for gender differences in the prevalence of these disorders (3-fold higher in males than females). In order to explore the hormonal basis for this phenomenon, rat pups were gonadectomized at postnatal day (PND) 14, 20, and 28, and D1 and D2 receptor density analyzed with autoradiography. Gonadectomy did not have a significant effect on either receptor type at PND40. In contrast, increased testosterone given to intact animals altered D1 dopamine receptor density in the striatum at trend level ($F_{1,26} = 3.32, p = 0.08$). In both males and females, D1 receptor density was elevated in striatum. The effect was not enduring, however, and by PND80, no significant difference was observed. For both D1 and D2 receptors, age exerted significant effects on receptor density in a sex-dependent fashion: Age $\times$ Sex interaction; D1 $F_{1,26} = 35.7, p < 0.001$; D2 $F_{1,24} = 13.52, p < 0.001$). These data suggest that testosterone plays a partial role in the overproduction of dopamine receptors across the periadolescent period. This work is funded by RO1-43743 to MHT.

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INDIVIDUAL DIFFERENCES IN CARDIAC AND HPA REACTIVITY TO A PSYCHOLOGICAL STRESSOR. N. Tottenham, S. W. Parker, C. Liu, & M. R. Gunnar, University of Minnesota; Minneapolis, MN 55455; USA.

For decades, researchers have examined endocrine and heart rate measures in child research to assess hypothalamic-pituitary-adrenocortical and autonomic activity in response to stress. More recently, research has attempted to abstract the separate influences of the parasympathetic, through measures of Vagal Tone (VT), and sympathetic nervous systems, through measure of pre-ejection period (PEP), on the cardiac stress response. By analyzing both PEP and VT, work on adults has shown that some individuals increase heart rate by lifting or reducing VT, while others do so by increasing sympathetic tone, while still others may do both. Those who achieve similar increases in heart rate but do so primarily by sympathetic regulation are also the ones who are more emotionally anxious, elevate cortisol more, and experience more "stress-related" health problems. The current study examined the physiological correlates (heart rate, PEP, VT, and cortisol) of reactivity to a standard psychological stress test, in 8- and 9-year-old girls. Physiologic measures revealed individual differences in girls' responses to the test. Changes in PEP were examined in relation to VT and HR to identify children who change HR more with PEP modulation. Group differences existed in VT patterns across conditions and HR patterns. Furthermore, the different patterns of sympathetic, parasympathetic, and endocrine responses were related to temperamental differences reported by parents. Being the first to examine the relationship between such measures in children, this study further supports the importance of examining multiple systems simultaneously in developing populations.

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NEONATAL STIMULATION ELIMINATES A POPULATION LEVEL LEFT-BIAS IN HIPPOCAMPAL VOLUME. R. Tierney, T. Verstynen, & A. Tang, Department of Psychology, The University of New Mexico, Albuquerque, NM, 87131, USA.

In rodents it has been shown that early life stimulation can modulate functional asymmetries in spatial and auditory processing and in paw preference. The current study was designed to investigate whether this stimulation also affects anatomical asymmetry, specifically asymmetry in hippocampal volume. From postnatal day 1 to 21, experimental pups ($n = 18$) were exposed to an unfamiliar non-home environment for 3 min daily while control pups ($n = 18$) remained in the homecage. At 8 months of age, tissue slices were stained with a cresyl violet dye, and the volumes of the left and right hippocampi were measured. We found, in the control rats, a significant population level asymmetry in hippocampal volume ($p < .01$ one-tailed), with the left hippocampus being significantly greater than the right. We did not detect the same left bias in the experimental group ($p > .10$ one-tailed). Although there was a rightward shift in hippocampal asymmetry due to experimental manipulation, this change in asymmetry did not reach significance level of .05. The same results were replicated in a separate study using 18 rats. To our knowledge, these results offer the first evidence of hippocampal volumetric asymmetry in rats.

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PRENATAL ETHANOL EXPOSURE WITH VOLUNTARY CONSUMPTION DURING PREGNANCY: LACK OF EFFECTS ON SPATIAL LEARNING IN ALCOHOL-PREFERRING RATS. T. D. Tran, D. Kincaid, T. K. Li, & C. R. Goodlett, Department of Psychology, IUPUI, Indianapolis, IN 46202, USA.

High alcohol drinking (HAD I, HAD II) and alcohol-preferring (P) lines of rats, genetically selected for voluntary drinking of ethanol (EtOH) in adulthood, were used to determine whether free-access consumption of alcohol during pregnancy can induce place learning deficits in the exposed offspring. Four treatment conditions during pregnancy were used: (1) scheduled 1-hour access to free-choice 30% EtOH (v/v); (2) pair-fed matched control; (3) ad lib food and water control; and (4) 24-hour voluntary access to 10% EtOH (v/v) [used only with the HAD lines]. Prior to breeding, the rats were initially given 2–3 days of forced consumption, then established on the designated schedule of 2-bottle choice access to EtOH (either 10 or 30% v/v) and water for the remainder of the experiment. Access was ended just before mating, reinstated on gestational day (GD) 6 of pregnancy, and continued through GD 20. Weight-matched pregnant dams were assigned either to the scheduled access (30% EtOH and water) condition or to a scheduled access to an isovolemic, isocaloric sucrose solution and water. The ad lib controls were not exposed to EtOH at any time during the study. Offspring were tested for 6 days on either the standard place training or random location training in the Morris maze task. Place performance was significantly better than random performance for all groups, and the alcohol-exposed groups were not significantly impaired relative to controls. This first report evaluating effects resulting from voluntary drinking during pregnancy in alcohol-preferring lines found no effects on juvenile place learning. It appears that the levels of gestational exposure were not sufficient to impair development of spatial learning behavior. Supported by NIAAA Grant #AA07611-13.

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EFFECT OF ETHANOL CONCENTRATION AND AGE ON SELF-ADMINISTRATION OF ETHANOL IN INFANT RAT. E. M. Truxell, & N. E. Spear, Center for Developmental Psychobiology, Binghamton University, Binghamton, NY, 1390, USA.

In order to determine the optimal concentration of ethanol for consumption by infant rats, a self-administration paradigm similar to that of Hall and Bryan (1981) was employed. Infant rats 8, 12, or 16 days old were placed into a glass terrarium in which the floor had been covered with a small (30 ml) amount of ethanol. After being voided of urine and fecal matter the pups were weighed. One male and one female pup from each litter were then placed into a chamber containing 0, 5, 10, 15, or 20%, ethanol, for a period of 20 min. After a 48-hour interval this process was repeated except that only water was present in all chambers. Following each session, pups were dried and weighed again. The difference in weight was calculated as percentage body weight increase. Absolute alcohol intake was determined in terms of grams ethanol per kilogram body weight (g/kg). A $2 \times 3 \times 5$ (sex, age, concentration) ANOVA on the ethanol day indicated main effects of both age and concentration. The same analysis on the water day revealed a main effect of age. These results confirm previous findings of greater intake of ethanol than water and add surprising intake of high concentrations (2.5–3/0 g/kg intake of 15 and 20%), exceeding that of 5% or water. Prior intake of ethanol tended to enhance water intake during the second session, especially in the 12-day-olds, and could indicate a reinforcing effect of ethanol.

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EARLY ENVIRONMENTAL ENRICHMENT: EFFECTS ON STEREOTYPED BEHAVIOR AND NEURONAL ACTIVITY. C. Turner, & M. Lewis, Departments of Psychiatry, Neuroscience & Psychology University of Florida, Gainesville, FL 32610, USA.

Environmental enrichment results in improved learning and various neuronal changes (e.g., increased dendritic arborization). Enrichment has also been shown to be neuroprotective in models of CNS insult (e.g., seizures). We have shown that early environmental enrichment prevents the development of stereotyped behaviors. This effect is seen in deer mice which develop spontaneous stereotypies when housed in standard cages. The purpose of this study was to determine regional brain changes associated with the enrichment-related decrease in stereotypy. To identify regional CNS changes associated with enrichment, cytochrome oxidase (CO) immunohistochemistry was used as a marker of neuronal metabolic activity. Mice (25d) were housed for 120 days in enriched environments and their stereotypic behavior assessed. Enrichment prevented the development of stereotypy in about 60% of the mice. Enriched mice exhibited significantly higher levels of CO staining than non-enriched mice in several striatal areas, nucleus accumbens and hippocampus. Furthermore, the enriched non-stereotypic mice exhibited higher levels of CO activity than either the enriched but stereotypic mice or standard cage (stereotypic) controls, which did not differ in CO activity. This effect was most pronounced in the frontal cortex and dorsal lateral striatum. Thus, environmental enrichment increased neuronal metabolic activity in discrete brain regions. Moreover, enriched mice not developing stereotypy exhibited increased CO activity in specific brain areas compared to stereotypic mice, independent of housing condition.

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SPONTANEOUS MOTILITY IN THE EMBRYONIC CHICK: RELATIONSHIPS BETWEEN THE DURATION OF QUIESCENT INTERVALS AND ACTIVITY EPISODES. L. B. Tucker,² J. Tabak,¹ & M. J. O'Donovan,¹ ¹Laboratory of Neural Control, NINDS-NIH, Bethesda, MD and ²Department of Psychology, University of Florida, Gainesville, FL, USA.

Episodic, spontaneous motor activity is observed in chicks beginning around embryonic day (E) 4. Furthermore, when the spinal cord of the embryo is isolated and maintained *in vitro*, it generates episodes of spontaneous activity that are similar to the electromyographic (EMG) activity recorded *in ovo* from hindlimb muscles. Results obtained from *in vitro* recordings have demonstrated a positive correlation between the duration of an episode of activity and the duration of the preceding, but not the following, inter-episode interval. These results suggest that slow activity-dependent network depression may be important in regulating the episodic, spontaneous activity *in vitro*. Because spontaneous activity *in vitro* is similar to EMG activity recorded *in ovo*, recovery from depression may also be a controlling factor during spontaneous motility *in ovo*.

To test this idea, we recorded spontaneous motility and EMG activity from three chick embryos *in ovo* to determine if the relationships between the duration of episodes and inter-episode intervals were similar to those found *in vitro*. For this purpose, we observed the spontaneous movements of White-Leghorn chicks *in ovo* for about 1 hr at 9. At the same time, EMG activity was recorded from the posterior iliotibialis muscle, using flexible suction electrodes. No significant positive correlations were found between the duration of activity episodes recorded either by observations or by EMG and the durations of the preceding, or the following inter-episode intervals. These preliminary results, if confirmed, suggest that the mechanisms hypothesized to regulate episode duration *in vitro*, such as synaptic depression, may not operate in the same way *in ovo*. Alternatively, the duration of movement episodes and single muscle EMG activity may not accurately reflect the excitability of spinal networks *in ovo*.

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ACUTE EFFECTS OF ETHANOL ON BEHAVIOR OF ADOLESCENT RATS: ROLE OF SOCIAL CONTEXT. E. I. Varlinskaya, L. P. Spear, & N. E. Spear, Center for Developmental Psychobiology, Binghamton University—SUNY, Binghamton, NY, USA.

The important property of alcohol that causes drinking in adolescents is its ability to enhance interpersonal effectiveness in social situations by facilitating certain forms of social behavior, particularly interactions with peers. The present study (1) characterized acute effects of alcohol on social behavior and social motivation (preference/avoidance of a peer) in adolescent rats and (2) compared ethanol effects on behavior directed either toward an inanimate novel object or toward a peer. In the social context the effects of ethanol were dose-dependent and biphasic. Low doses of ethanol (0.25–0.75 g/kg) produced apparent social facilitation (increased social activity and enhanced social preference) while higher doses (3 and 4 g/kg) were effective in causing social inhibition (decreased social activity and avoidance of a peer). This pattern was not observed in a nonsocial stimulus: though overall activity in the nonsocial context was suppressed by 2 and 3 g/kg of ethanol, 0.5 g/kg of ethanol produced no activating effects on overall activity directed to the inanimate object. Taken together these results suggest that the social nature of the testing situation plays an important role in responsiveness to alcohol in adolescence. This research was supported by grants from the National Institute of Alcohol Abuse and Alcoholism (AA12525 and AA12150) to L.P. Spear and (AA11960) to N.E. Spear.

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SOCIAL EXPERIENCE OF FLOCKMATES INFLUENCES DEVELOPMENT OF COURTSHIP AND COMMUNICATION IN COWBIRDS, *MOLOTHRUS ATER*. D. J. White, M. J. West, & A. P. King, Department of Psychology, Indiana University, Bloomington, IN 47405, USA.

In a diversity of species, individuals depend on interactions with others to learn a vast amount of their behavioral repertoire. The impact of the social environment on development of mating behavior was investigated in cowbirds. Using naturally occurring variation in social experience, distinct social environments were created in large aviaries for male and female cowbirds. Groups of juvenile and adult females were housed with: (1) juvenile and adult males, (2) only juvenile males, (3) only adult males, or (4) no males. Across the four conditions, dramatically different cultures emerged from the differences in social composition. Males across conditions differed in: amount, quality, and type of song produced as well as the ways in which they used their song. Males across conditions also exhibited differences in aggression, mate guarding, and pair bonding. Furthermore, both males and females differed across conditions in how they assorted spatially. Results are consistent with the hypothesis that the social environment can impact the developmental trajectory of mating behavior.

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ALCOHOL IMPAIRS BOTH ACQUISITION AND EXPRESSION OF PLACE LEARNING IN ADOLESCENT RATS. J. Wilber, & C. R. Goodlett, Department of Psychology, Indianapolis, IN 46202-3275, USA.

In the Morris water maze spatial navigation task, adolescent rats given alcohol before daily training sessions show acquisition deficits in place learning, at doses that do not impair adults. The current study tested whether alcohol could also impair expression of a place discrimination that was recently learned by adolescent rats, and evaluated whether state-dependent learning effects could be produced. On postnatal day 26, Long-Evans rats were assigned to saline or alcohol (2.0 g/kg) treatment groups, and given injections 20 min before the start of each daily Morris maze training session. For place training, a submerged platform remained in the same location on all trials. Training ensued for 6 consecutive days, 4 trials per day. On the 7th day, rats of each treatment group were given either saline or alcohol, and tested 20–30 min later on a single probe trial, with the escape platform removed. Training controls included a visible platform condition, and a random submerged condition. Alcohol during acquisition selectively impaired (but did not fully prevent) place learning, and eliminated expression of target place biases on the probe trial, even when tested under saline. Alcohol also impaired probe performance of the groups given acquisition training under saline. Alcohol had no effect on acquisition performance in the visible or random platform conditions. In adolescent rats, alcohol-induced deficits in spatial behavior may reflect acute disruption of the neural coding or representation of space, disrupting both the acquisition and the expression of place learning.

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CHANGES IN STRESS-INDUCED C-FOS EXPRESSION DURING EARLY ONTOGENY IN RATS. C. P. Wiedenmayer,^{1,3} & G. A. Barr,^{1,2,3} ¹Department of Psychiatry Columbia University, ²Department of Psychology, Hunter College, and ³Development of Psychobiology, NY State Psychiatric Institute, New York, NY 10032, USA.

When exposed to a potentially infanticidal adult male, young rats become immobile. Male-induced immobility declines during the preweaning period, paralleling the decrease in infanticidal threat. To investigate the neural substrates underlying the developmental change in immobility, male-induced expression of the immediate-early gene *c-fos* was assessed. On postnatal days 14 and 21, a huddle of three male rats was exposed to an adult male behind a screen. As control, three littermates were put in the testing chamber but not exposed to the male. Two hours after testing, the rats were sacrificed, their brains removed and processed for immunocytochemistry. On Day 14, rats in the presence of the male stopped ongoing behaviors and became immobile. They had significantly higher *c-fos* expression in the paraventricular nucleus of the hypothalamus, the amygdala, the ventral tegmental area, the periaqueductal gray, and the locus ceruleus. On Day 21, male-exposed rats that were more immobile than controls expressed *c-fos* in a similar pattern as 14-day-old rats; however, different nuclei of the amygdala were activated. In contrast, male-exposed 21-day-old rats that showed no increases in immobility did not have elevated *c-fos* expression in these areas. Positive correlation between number of labeled cells and immobility indicates that the periaqueductal gray and the locus ceruleus contribute to immobility. These results demonstrate that male exposure activates brain areas of young rats in an age-specific way. Supported by NS36130, DA00325 and NS10817.

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LESIONS OF THE SUPERIOR COLLICULUS AND VENTRAL MEDIAL THALAMUS ATTENUATE U-50,488-INDUCED LOCOMOTOR ACTIVITY OF YOUNG RATS. A. R. Zavala, J. R. Osburn, S. T. Yoshida, & S. A. McDougall, Department of Psychology, California State University, San Bernardino, CA 92407, USA.

Systemic injection of a κ -opioid agonist (e.g., U-50,488) increases the locomotor activity of preweanling rats. κ -Opioid receptors in the substantia nigra pars reticulata mediate this effect, because microinjecting U-50,488 into this structure induces locomotion in 18-day-old rats. The purpose of the present study was to determine whether the nondopaminergic nigroreticular and nigrothalamic output pathways mediate U-50,488-induced locomotor activity. Therefore, separate groups of rats were given bilateral sham or electrolytic lesions of the superior colliculus (SC; 1 mA/55 s) or ventral medial thalamus (TVM; 1 mA/20 s) on postnatal day (PND) 16. On PND 18, rats were placed in testing chambers and locomotor activity was measured. After 15 min of behavioral assessment, rats were injected with saline or U-50,488 (5.0 mg/kg, ip) and returned to the chambers for an additional 45 min. On the next day (i.e., PND 19), the identical procedure was employed except that rats were injected with saline or the direct D₂-like receptor agonist NPA (0.01 mg/kg, IP). Results showed that U-50,488 increased the locomotor activity of preweanling rats. This effect was partially attenuated by bilateral lesions of either the SC or TVM. DA agonist-induced locomotor activity was not affected by lesions of the SC or TVM. Thus, these results show that the nigroreticular and nigrothalamic pathways mediate, at least partially, the locomotor activity induced by κ -opioid receptor stimulation, but that these same pathways do not mediate locomotion induced by D₂-like receptor stimulation.

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CAN MOM TELL WHETHER YOU'VE LEFT HOME?—EVIDENCE FROM MATERNAL OBSERVATIONS OF THE RAT. B. Zou, B. Reeb, T. Verstynen, & A. Tang, Department of Psychology, The University of New Mexico, Albuquerque, NM 87131, USA.

Our previous study has shown that maternal behavior can interact with neonatal stimulation to modulate learning performance in the rat. This interaction effect could be mediated through differential maternal behavior toward the stimulated (S) and non-stimulated (NS) pups. In this study, we investigated whether a dam indeed displays differential behavior toward the S and NS littermates. Forty-eight pups from six litters were used, with halves of each litter assigned to the experimental and control group,

respectively. From postnatal day 1 to 21, the dam was first removed from the home cage and the experimental pups were exposed to an unfamiliar non-home environment for 3 min while control pups remained in the home cage. Both groups were touched in an equal amount by the experimenter. Upon returning to the home cage, experimental and control groups were placed in two separate piles within the home cage before the dam was returned. Analysis of the post-stimulation dam-pup interaction revealed that the dam's retrieval latency was significantly shorter for the experimental than for the control pups ($p < .05$ one-tailed). This result demonstrates that a rat dam can discriminate, among her own litter, those pups briefly exposed to an unfamiliar environment and those that were not. This differential treatment suggests that prompt maternal care may facilitate the neonatal stimulation-induced learning enhancement through an earlier termination of the stress and through other subsequent preferential treatment.